Structures 80 (2025) 110115

Contents lists available at ScienceDirect

Structures

journal homepage: www.elsevier.com/locate/structures

Methodological framework for integrating structural health monitoring and digital-twin models for seismic assessment of heritage buildings: Case study of Basilica Maria Auxiliadora, Lima, Peru

Hector Aroquipa [a,b,*], Alvaro Hurtado [a,b], Christiam Angel [c]

[a] Research Group on Numerical and Computational Methods, Graphic and Scientific Computing, Instituto de Investigación Científica, Universidad de Lima, Av. Javier Prado Este 4600, Santiago de Surco 15023, Peru
[b] Civil Engineering School, Universidad Nacional Autónoma de Tayacaja Daniel Hernández Morillo, Tayacaja, Huancavelica, Peru
[c] School of Architecture, Urban Planning and Construction Engineering, Politecnico di Milano, Milan, Italy

ARTICLE INFO

Keywords:
FEM numerical models
Seismic performance evaluation
Intelligent circular resilience
Heritage buildings
Digital twin
Structural health monitoring

ABSTRACT

This study proposes a methodological framework for integrating structural health monitoring (SHM) systems with numerical digital twin (DT) models for the seismic assessment of heritage structures. This approach is being implemented within smart cities, focusing on intelligent circular resilience. The process begins in the physical world by recording data through strategically placed sensors distributed throughout the evaluated structures. Afterwards, the virtual world refers to the development of numerical models that capture the structural performance of the building while considering specific features to simulate the original one. The interconnection of both sides requires exhaustive and detailed cross-validation to ensure an established level of reliability, often based on the expected accuracy and limitations. Nevertheless, extending this process to heritage buildings demands a more sophisticated approach, designed to identify and classify discrete damage levels, and involves establishing proper threshold limits for an early warning/alarm system. This paper presents a methodology focused on the virtual implementation of these processes, aiming to integrate such technologies effectively. The case study is a religious monument that has been standing for over a century, located in a high seismic hazard zone in Lima, Peru. Numerical models were developed using 3DEC and SAP2000 software to capture the building's dynamic response, including modal and time-history analyses for a set of 34 seismic ground motion records. The results were used to identify susceptible zones and the location of sensors. Further recommendations address the challenges encountered in implementing this approach and outline the next steps, including sensor deployment and validation against experimental models.

1. Introduction

In modern engineering, the characterization and estimation of both the seismic response and the performance of structural and non-structural systems have evolved through the incorporation of innovative technologies. These tools allow users to capture and integrate all building components into a virtual model through numerical models, which are meant to approximate the behavior of structures and on uncertainties on the level and frequency of extreme loadings. However, these representations are constrained by the scope of work, the reliability required for each particular project, the level of detailing of the model, and the capabilities/limitations of the software for the simulation. In addition, access to specific building features related to geometry, material properties, damage reports from past events, and interventions plays a key role when determining the model's adequacy and accuracy in properly capturing the seismic response of the construction. Nevertheless, this practice has nowadays been widely standardized, to some extent, in national and international codes and standards for common practice [1,2], under what has been cataloged as seismic vulnerability assessment for existing buildings.

Special attention has been given to buildings classified as heritage buildings, which present a unique structural behavior when subjected to external loadings. In fact, many seismic vulnerability studies have demonstrated that, despite having similar features from a geometric and material standpoint, their seismic response cannot be grouped into a generic category [3–7]. Moreover, features related to construction

* Correspondence to: Universidad Nacional Autónoma de Tayacaja, Tayacaja, Huancavelica, Peru.
E-mail address: haroquipa@unap.edu.pe (H. Aroquipa).

https://doi.org/10.1016/j.istruc.2025.110115
Received 23 April 2025; Received in revised form 6 August 2025; Accepted 29 August 2025
Available online 8 September 2025
2352-0124/© 2025 Institution of Structural Engineers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

techniques utilized during their construction differ from one region to another, increasing the uncertainty of sources [8,9]. As a result to overcome this challenge, a joint effort has been made by professionals and academia to outline guidance on how to address these kinds of challenges [6,10,11], and the limitations that arise when characterizing construction materials due to restrictions on performing destructive tests [12–17].

Despite all these challenges faced by structural engineers in characterizing the seismic response of heritage buildings, major research projects have been developed worldwide to preserve their historical integrity [18,19], leading to the development of alternative techniques integrating technologies that feed traditional numerical modelling. Nowadays, Structural Health Monitoring (SHM) is one of them, focused on capturing the building's state by installing sensors to measure real-time parameters [20–24]. The sensor locations are based on expert judgement and results from Finite-Element Method (FEM) simulations, which focus on components highly susceptible to seismic actions [21,25–27]. One of the main goals of SHM is the creation of systems capable of predicting damage levels. In this context, the digital revolution 4.0 offers a new solution called digital twin (DT) models [28–32], meant to connect the real and virtual worlds by creating a digital model that represents the actual asset, while in the physical world, there are sensors placed strategically, streamlining the real-time flow of information [33–36].

DT model integration within SHM systems has been addressed by several researchers [37–42], which agrees on the fact that these may be

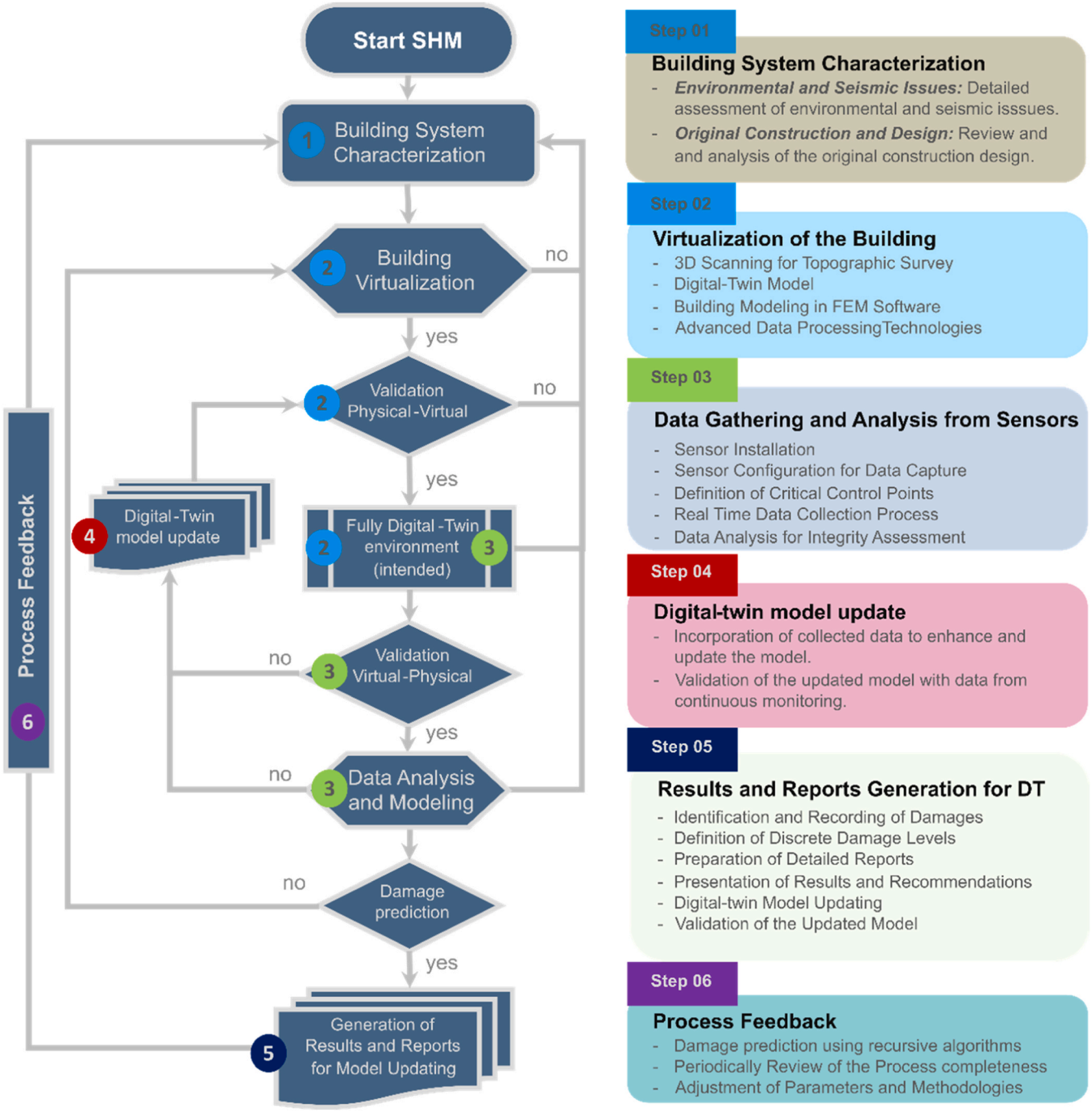


Fig. 1. Overall algorithm of the proposed methodological approach.[58]

extended to smart cities and disaster risk reduction (DRR) management. At this point, the Intelligent Circular Resilience (ICR) is proposed as a novel methodological framework aimed at improving DRR management by extending the concept of dynamic circular seismic resilience through the integration of SHM, real-time data analysis, and active public-private participation. The framework operates across three stages—understanding seismic risk, implementing mitigation strategies, and responding to seismic events—within a continuous, circular process that identifies blind spots in risk perception, institutional coordination, and infrastructure planning. ICR is designed to function on three interdependent levels: institutional expertise, infrastructure risk management, and real-time monitoring, each supported by the adoption of emerging technologies toward the development of intelligent and resilient urban environments [43].

As a result, the authors proposed a novel methodological approach focused on dynamic ICR- refer to Fig. 1, which emphasizes the development of DT models from the virtual world standpoint. On one hand, this approach addresses the challenges and limitations encountered in heritage buildings by identifying critical features such as geometry configuration, material properties, and available historical records. On the other hand, it deals with the duality of the physical and virtual world through the selection and installation of sensor devices and the development of a trustworthy DT model for continuous feedback. This research delves into the Building Virtualization and partially on Data Gathering and Analysis from Sensors phases outlined in Fig. 1(i.e., steps 02 and 03, respectively). The broad development of the methodology will be complete in further publications, incorporating experimental models and sensor deployment, covering phases 3–6.

In this study, the DT is defined as a dynamic numerical representation of the heritage building that replicates its behavior under seismic loading, continuously updated and cross-validated through real-time data from strategically placed sensors. This virtual replica integrates geometric fidelity, material heterogeneity, historical performance data, and predictive deep learning (DL) algorithms, enabling proactive structural assessment and decision-making. Geometrical accuracy is critical in the development of DT models, especially when dealing with heritage buildings whose non-linear seismic behavior is highly sensitive to local geometric irregularities. Recent studies (e.g., [44,45]) demonstrate the power of combining 3D scanning technologies and photogrammetric methods to reconstruct detailed digital geometries suitable for structural analysis. In parallel, recent work by [46] outlines a systematic pipeline for generating discrete element models (DEM) directly from imagery, reinforcing the potential of image-to-DEM workflows for data-driven seismic assessment.

The analyzed phases were achieved through three main steps; the first provides guidance on the available and recommended field survey methods to characterize the geometric configuration and material properties using a Building Information Model (BIM). The second focuses on the creation of the virtual model using a finite element method (FEM) model and covers key aspects such as software selection based on its features, capabilities, and limitations; the linking process between the BIM and FEM model, modeling techniques, and seismic hazard definition are also well-thought-out. Finally, the last phase highlights the challenges that arise during the post-processing of results and highlights the available tools to analyze massive data by suggesting a novel idea that incorporates the intensity-duration-frequency plots for the engineering demand parameters (EDP). Therefore, all these steps lead to the integration process between the physical and virtual world under the ICR framework (see Fig. 2).

The process deals with aspects related to the deployment of a reliable SHM while focusing on the Early-warning system (EWS) and its connectivity between both worlds in real-time. Specialized sensor devices installed on the building will cover the physical world mapping response, while a FEM model is meant to represent the DT model in the virtual world. Thus, sensor software and hardware requirements, as well as the location selection for installing those, are defined using the performance-based assessment from a non-linear FEM model. In addition, these devices measure movement in terms of displacement,

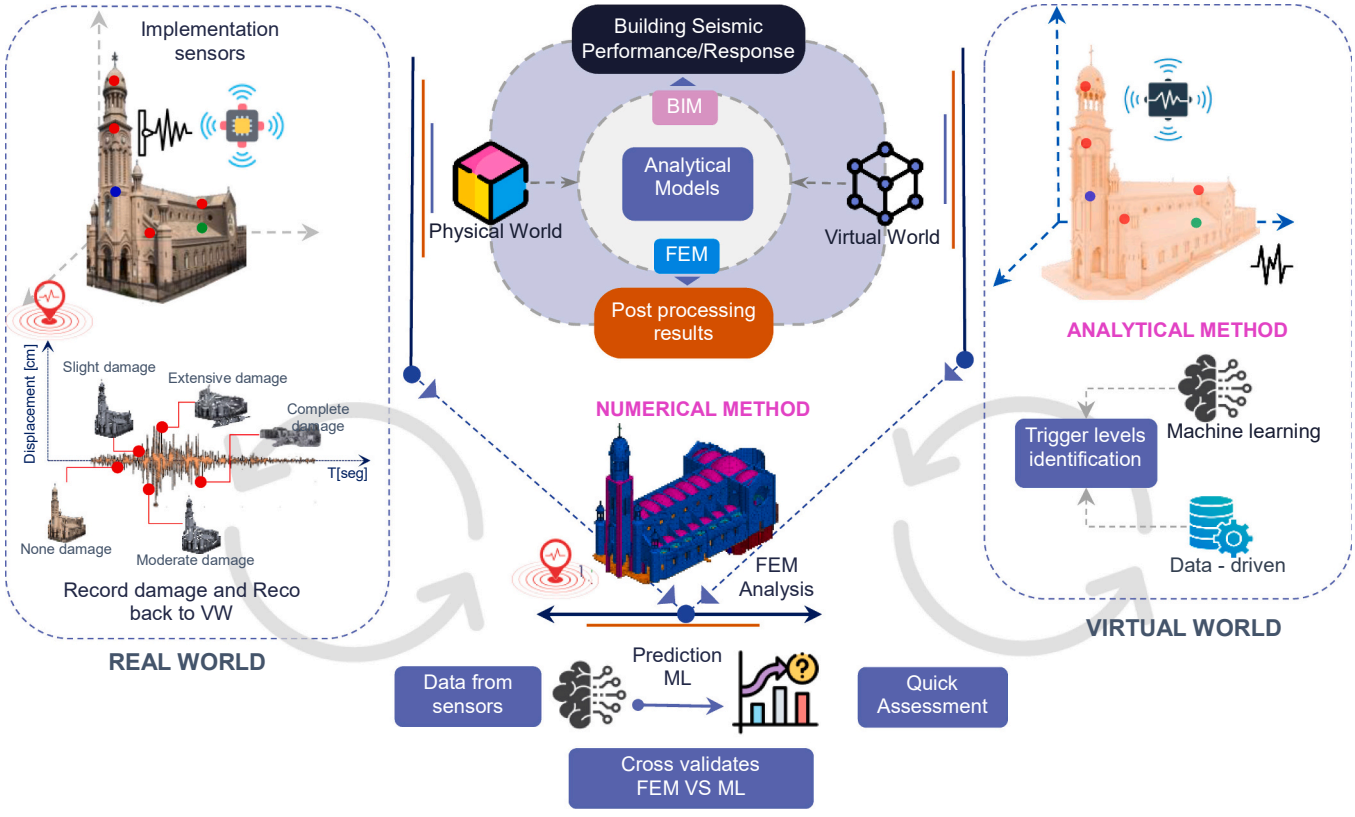


Fig. 2. Conceptual framework of the digital twin integration process, based on FEM modeling for both virtual representation and physical feedback simulation.[58]

velocities, or accelerations in bi- or tri-axial directions. Hence, the linking process to identify damage in structural components (e.g., masonry walls) using the data gathered by them requires employing transfer functions (TF) obtained through DL algorithms. However, a primer analysis of the TF uses the virtual world results due to the lack of existing recorded data, which can further be calibrated via the new data gathered in the physical world. That creates a feedback procedure between both worlds and its DT model.

Recent developments in artificial intelligence have introduced advanced techniques for crack detection using DL and computer vision, with proven applications in masonry and concrete structures. Studies such as [47] and [48] have demonstrated the potential of convolutional neural networks (CNNs) for accurate, real-time crack localization and quantification. Although this study focuses on numerical and sensor-based modeling, these techniques are highly relevant and may be integrated into the early damage identification process or serve as validation tools for physical damage post-event.

The case study corresponds to a heritage building located in Lima, Peru, specifically a religious monument that was built a century ago**.** A field survey was conducted to portray the geometric configuration through a point-cloud survey using a 3D laser scanner (3D LiDAR technology) that was then converted into a BIM, while the definition of the mechanical properties of the materials was based on previous studies and archives. In addition, detailed research into the building's performance during previous earthquakes was addressed by checking existing damage reports, including the 2007 Pisco earthquake (moment magnitude, Mw = 8.0). Subsequently, the virtual model was developed using the software 3DEC Version 9.0 [49], based on an intermediate calibration and validation process, which consisted of approximately 4500 block elements and 161,000 contact zones, and analyzed a total of 5 seismic records through time-history analysis (THA). The results were used to identify the monument's seismic response, compare it with the previous damage report, and define the best spots for installing the field sensors that would be integrated into the physical world via the DT model.

The main novelty of this research lies in the development of a methodology integrated with the ICR framework for heritage buildings. This methodology leverages SHM through a DT model, combined with a numerical one that focuses on the virtualization of the building. The technical aspects are addressed using a performance-based assessment, which captures the seismic structural response through a novel way of using the intensity-duration-frequency (IDF) for characterizing EDPs' results. This approach condenses the vast data generated from the THA results, allowing the identification of critical distribution patterns for the selection, placement, and calibration of sensors on the building, as well as TF for estimating damage on masonry walls. Finally, the methodology proposes action strategies to overcome the challenges encountered throughout the entire implementation process. Furthermore, it can be extended to other heritage buildings.

2. Characteristics of the heritage building: Maria Auxiliadora Basilica

A religious monument was selected as the case study heritage building: the Maria Auxiliadora Basilica, located in the Breña district of the city of Lima, Peru (i.e., a very-high seismic hazard zone, refer to Section 3.2). The construction process lasted around 8 years, from May 1916 to March 1924, which was initially labeled as a sanctuary until 1962, when it was recognized as a basilica. Fig. 3 shows some pictures of the monument, while a detailed description of its main features can be found in [50]. Since its construction, the building has not undergone major modifications to its original structure, except for two interventions due to restoration works after specific earthquake events.

The main features related to the geometric configuration and construction materials (i.e., brick masonry units, mortar, concrete, steel, and stone masonry for the foundations) were obtained from the Maria Auxiliadora archives [50], which also provided some useful insights regarding the construction process and techniques. These records clarified many of the connections between different materials, such as truss support on masonry walls and encasement of steel profiles in the main columns, details that are sometimes difficult to identify through field surveys without performing destructive explorations. As a result, all the information provides valuable insight into the DT model by enhancing the accuracy in the representation of the physical world in the virtual one and reducing the uncertainty related to building configuration and construction techniques lost over time, as evidenced in other studies [51–54]. This translates into a more reliable numerical model, increasing the accuracy of the structural representation and its interaction with the different materials comprising the building. The next subsections present the characteristics of the mechanical material properties, geometric configurations, and a summary of the performance from previous earthquake events that were used to calibrate FEM models.

2.1. Materials

The heritage building comprises several materials, including masonry, concrete, steel, and mortar. The foundation system consists of masonry with clay brick units and cement mortar joints. For the columns of the nave, some utilize concrete-encased steel H-shaped profiles, while all external and internal walls are made of masonry finished with a concrete mortar layer. The remaining columns are unreinforced brick masonry, confirmed through architectural plans and visual inspection. As for the domes, they were built using several layered materials, including a reinforced concrete slab and brick arches. Furthermore, the roof consists of cast iron trusses with L-shapes, holding the clayey shingles that are supported on the perimeter walls. The connections between steel elements were built using rivets, as this was the latest technique at the time of construction.

Direct field characterization through destructive testing was not possible due to preservation regulations for heritage buildings. While non-destructive dynamic techniques such as ambient vibration testing (AVT) are viable alternatives (see [55]), the present study relied on archival and prior technical data due to the scope of this methodological phase. AVT-based calibration will be considered in future work during the sensor integration phase. Table 1 summarizes the mechanical properties of the materials used for the FEM model, which were derived from previous studies and detailed research in archives [56]. That highlights one of the main challenges faced when assessing heritage buildings [8,15,17,24,57], which forces engineers to rely on alternative sources or similar buildings built during the same period [15].

2.2. Geometry

The geometric characterization of the *Basilica María Auxiliadora* was conducted using state-of-the-art terrestrial LiDAR technology to capture the macro-geometric configuration of the monument with the highest possible spatial fidelity. A high-density 3D laser scanning system was employed, capable of acquiring up to 976,000 points per second, with an effective operating range from 0.6 m to 120 m, suitable for both interior environments with low ambient lighting and reflective exterior surfaces. The system offers a maximum distance measurement error of ±2 mm at 100 m, validated for surfaces with both high and low reflectivity, ensuring accuracy even under complex architectural and ornamental conditions typical of historical structures.

The scanner integrates a dual-axis compensator with an angular precision of 0.015°, allowing automatic correction of leveling deviations during field acquisition. Additionally, an electronic barometric height sensor and digital compass provided georeferenced orientation and vertical positioning for each scan station, facilitating global alignment and registration of the point cloud dataset. The device also includes 70-megapixel color resolution, enabling the generation of textured and

Fig. 3. Pictures of the monument: a) old record photograph from 1929, and b) current photograph from 2023 (source: Salesian community archive record).

colorized point clouds, an essential input for accurate 3D modeling and heritage documentation.

Based on the processed point cloud, illustrated in Fig. 4—a BIM was developed, which was subsequently validated through an exhaustive comparison with the topographic records, following the methodology detailed in Fig. 1. This verification confirmed the unaltered spatial layout of the original architectural configuration since the building's construction, as documented in historical archives and the original design plans by the Italian architect Ernesto Vespignani [50,56]. This workflow ensured a dimensionally accurate and verifiable geometric base for numerical modeling, establishing a robust foundation for subsequent structural assessments under both static and dynamic loading conditions.

The building exhibits the typical spatial organization of religious architecture in the Byzantine-Romanesque style, including a single-cuspid front, central nave, lateral aisles, presbytery, side chapels, a subterranean crypt, and an auxiliary service area located on the east side (see Fig. 5). The overall footprint of the structure spans approximately 2000 m^2, with dimensions of 70 m in length, 30 m in width, and a front tower rising 56 m.

Subsequently, the final geometric configuration used for the FEM model creation is depicted in Fig. 6, aiming to illustrate some of the main dimensions, while Table 2 summarizes the typical cross-sections for the most relevant structural elements. The main structural system is comprised of masonry bearing walls on the façade, while the inner walls are supported on reinforced concrete columns and masonry arches that interconnect with the outer walls. Likewise, the roof is comprised of a steel truss supported on the load-bearing walls as well as the reinforced concrete vaults.

2.3. Building performance from previous earthquakes

Extensive research was conducted to understand the building's behavior during past earthquake events by consulting existing damage record reports. A detailed repository with records of the main events affecting the monument (held by the Salesian community) was reviewed to establish the date of each record and obtain a brief description. From the whole, a total of 5 earthquake-related episodes were relevant [58]. One of these events was the earthquake on May 25th, 1940, with its epicenter in Lima and a moment magnitude of Mw = 7.90. Furthermore, an intervention process was performed in 2012 due to damage sustained by the Ica earthquake (Mw = 7.90) on August 15th, 2007, which caused considerable damage to the building components, as indicated in the report prepared by "Universidad Nacional de Ingeniería" (UNI-CISMID) and "Pontificia Universidad Católica del Perú" (PUCP) in 2009 [50]. A summary of the effects and the retrofitting campaign is presented in [59]. From the author's side, a technical analysis was conducted by revising historical photographs [59], which allowed identifying minor crack patterns throughout various components ranging from 0.01 mm to 4 mm. However, there was a particular element with a 7 mm crack width. The main findings are summarized below:

- Apse: vertical cracks between the wall joints and minor cracks on vaults.
- Central tower: vertical cracks in the central tower and west façade.
- Main access arch at central tower: vertical cracks in the main access arch.
- Lateral left tower: vertical and horizontal cracks, with no clear patterns.
- Exterior vault: cracks at joints and longitudinal minor cracks along its perimeter.
- Interior vault: cracks at joints between vaults' arches and minor cracks in the longitudinal vault in the main nave.
- Main columns: cracks concentrated at the column base up to the first meter and spreading above with no clear pattern.

3. Seismic demand

3.1. Seismic record selection

A complete set of seismic records was selected to conduct the non-linear THA of the monument. The process consisted of filtering events located within a spatial range from the building within a specific time window; the range was established as a radius of influence of 300 km, while the window considered the time lapses between May 1916 and October 2021. Once the criterion was clear, the earthquake catalog database held by the Geophysical Institute of Peru, IGP (after its initials in Spanish "Instituto Geofísico del Perú"[1]), was revised to obtain a total of 5745 records. Subsequently, due to the amount of information, an

[1] https://ultimosismo.igp.gob.pe/datos-sismicos

Table 1
Summary of the mechanical properties for the different materials identified.

Parameter	Component		Value	Unit	Comment
Masonry unit type	Masonry		Brick	-	Dimensions: 12X6X25cm
Masonry material type			Isotropic	-	-
Masonry unit class			T - 1	-	According to NE−070
Masonry Unit Volume Weight	Masonry walls		17.40	kN/m^3	INF-LE335.08
Masonry unit compressive strength		f_b:	4.90	MPa	INF-LE335.08
Masonry walls		f_m:	4.53	MPa	INF-LE335.08
Mortar compressive strength		f_{cm}:	4.76	MPa	INF-LE335.08
Characteristic diagonal wall compressive strength**		f_k:	2.00	MPa	INF-LE335.08
Wall initial shear strength		f_v:	0.41	MPa	INF-LE335.08
Modulus of elasticity of Masonry		E_{wall}:	418.9	MPa	INF-LE335.08
Modulus of elasticity of Mortar		$E_{c\text{-}m}$:	5453	MPa	INF-LE335.08
Shear modulus of masonry		G_{wall}:	302.5	MPa	Eq.
Shear resistance of masonry		v_{wall}:	0.25	-	
Poisson ratio		ν:	0.72	-	
Masonry Coefficient of friction		u_{wall}:	0.75	-	
Elastic limit of steel		f_y:	210	MPa	According to NE−070
Modulus of elasticity of steel		E_s:	200,000	MPa	According to NE−070
Structural concrete, compressive strength		f_c:	17.5	MPa	According to NE−070
Modulus of elasticity of reinforced concrete		E_c:	16,315	MPa	According to NE−070

Notes:
*INF-LE335.08-Proyecto de Evaluación Estructural y reforzamiento de la Basílica María Auxiliadora.
** The value of fk corresponds to the characteristic compressive diagonal strength of masonry walls estimated from archival data and indirect methods, per NE-070 guidelines and INF-LE335.08

Fig. 4. 3D views from topographic survey results.

additional filtering process was applied, selecting only those records with a magnitude greater than or equal to Mw = 6.0, based on the fact that damage reports confirmed that lower intensities did not trigger any harm to the building; the shortlisting dropped the number to 75. Then, the time series were obtained from the seismic database prepared by the National Institute (responsible for recording seismic activity in Peru, called "Centro Peruano Japonés de Investigación Sísmica y Mitigación de Desastres" - CISMID[2]). It is important to note that, after accessing this portfolio, only 12 of the 75 had available information, meaning that a final list of 33 ground motions was used for the THA. Fig. 7 displays their location and magnitude, while Fig. 8 presents the input ground motion records selected for the THA, grouped by direction (N-W and E-W), to illustrate the variability in signal characteristics and durations. To enable a direct comparison with previous damage reports, the seismic ground motion records were analyzed without scaling or spectrum matching.

3.2. Seismic record characteristics

The seismic ground motion records were corrected using a linear baseline and Butterworth bandpass filter for a frequency range of 0.10–50 Hz. Next, the main characteristics of peak-ground acceleration (PGA) were analyzed and disaggregated by distance, magnitude, and events; refer to Fig. 9 for the disaggregation plot, and to Table A.1 for the complete set of characteristics of the chosen earthquakes. Finally, Fig. 10 depicts the comparison among the elastic spectral response spectrum (RS) for all components (i.e., E-W and N-S directions) compared against the design elastic spectral RS obtained from the Peruvian national standard code [60] for a soil type S_2, risk category 4, PGA of 0.45 g, and amplification factor Fa = 2.50. This figure suggests that none of the *RS* events overpass the design spectrum, except for one record (i.e., event record ID ICA002_20070815184058, *N-S* direction), which rises steeply at structural periods greater than 2.7 sec. Lastly, a 40 % reduction of elastic spectral acceleration for the range of constant acceleration can be observed, which then drops to 30 % for the range of constant velocities.

[2] http://www.cismid.uni.edu.pe/ceois/red

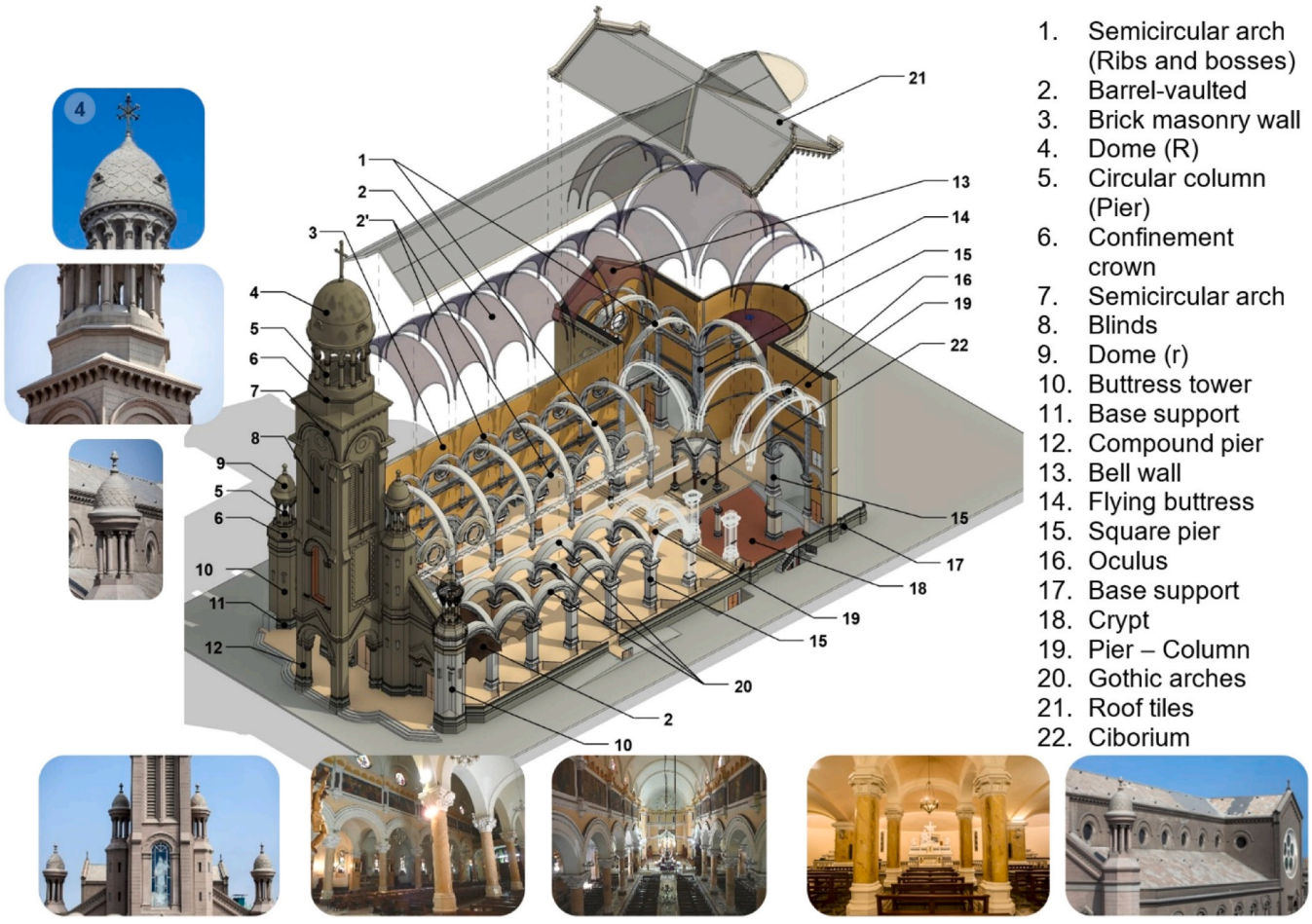


Fig. 5. Main architectural features of the monument.

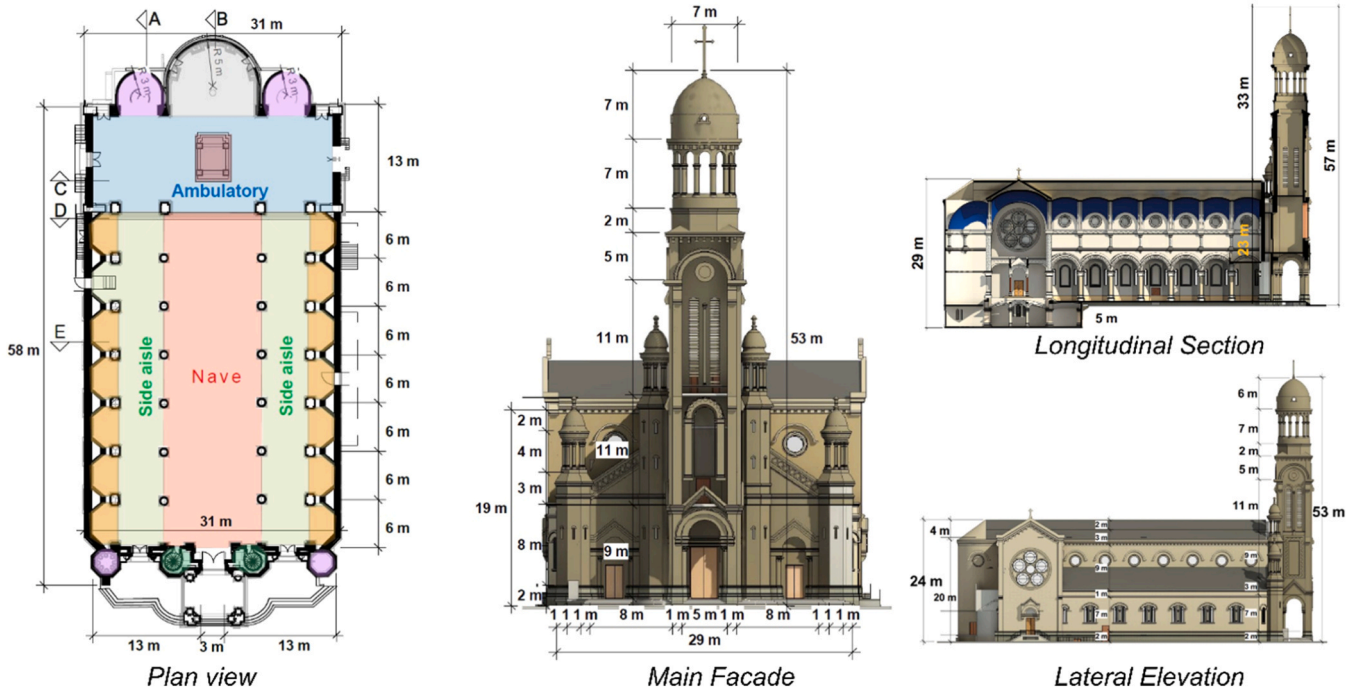


Fig. 6. Main geometric dimensions from the topographic survey results of the monument.

Table 2
Summary of the cross-section of the main structural components.

Parameter	Unit	Value	Comment
Masonry walls			
Type 1	m	0.80	External walls
Type 2	m	0.30	Outer and Inner walls
Reinforced concrete columns			
Type 1	m	0.60 × 0.80	Rectangular cross-section
Type 2	m	0.90 × 0.90	
Type 3	m	0.60	Circular cross-section
Barred vaulted			
Type 1	m	0.18	-
Type 2	m	0.28	-

4. Resilience and performance-based assessment

SHM and DT models were integrated through an EWS, which will be deployed in future research once the completion of methodology presented in Fig. 1 is completed. The inputs required for implementing the EWS follow a three-stage process, starting from a performance-based assessment of the heritage building, followed by defining the best location for installing the sensors. Lastly, the implementation of TF was performed to convert the movement that will be gathered by the sensors into structural damage on the masonry walls. The overall process is shown in Fig. 11. A detailed FEM model was developed considering the main findings depicted in the BIM, the expected non-linear response of all materials comprising the structural components, and the loading conditions for gravitational and lateral loads. Intermediate calibration was conducted to validate the results of the model, followed by analyzing the seismic response through 5 THA.

Then, the results from the FEM model were utilized as input to define the sensor position. During this process, several regions of interest (RoI) were defined to assess the most representative zone for installing them. Likewise, the IDF plots allowed for the identification of the range of operation and frequency sampling held in these devices. Finally, a set of TFs for each sensor was estimated using the Multiple-Input-Single-Output (MISO) approach and ML algorithm (see Eq. 1).

$$\sigma(t) = TF[u(t), \quad v(t), \quad a(t)] + \varepsilon \tag{1}$$

where either the acceleration $a(t)$, velocity $v(t)$, and displacement $u(t)$ time series results represent the input signals at time t, and the output signal is one of the main axial stresses on the masonry walls $\sigma(t)$, and ε denotes the prediction error. Additionally, an effective area of influence for each sensor was established to collect the input/output signals from the FEM model results.

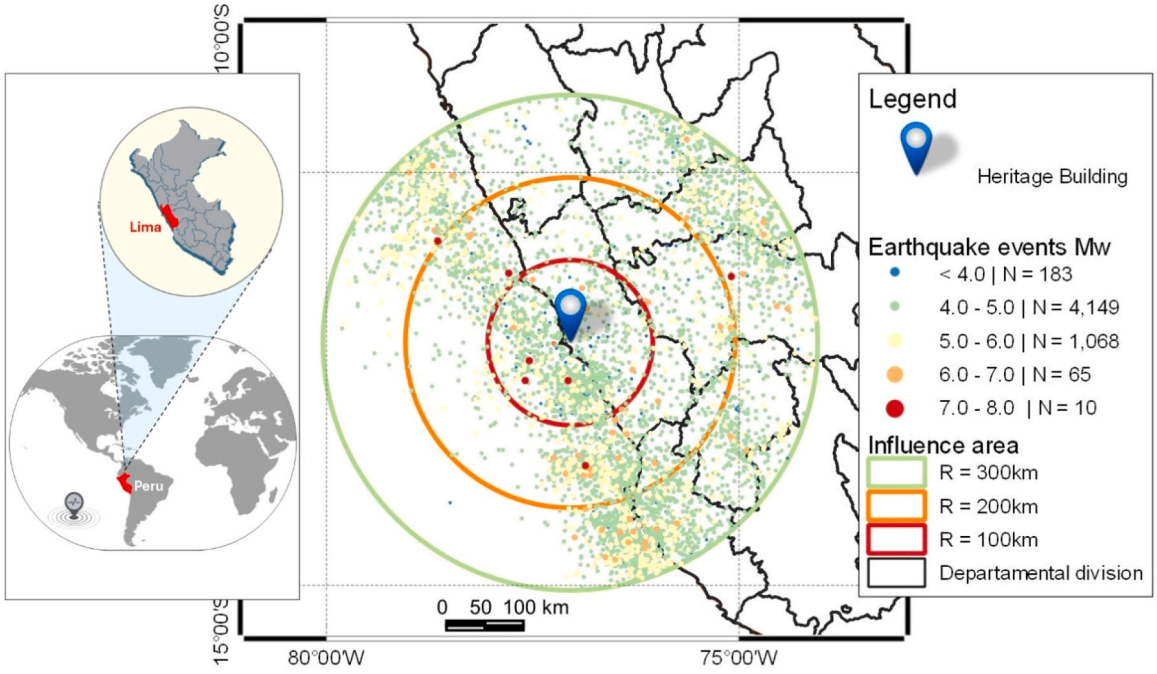


Fig. 7. Earthquake events selected for ground motion records.

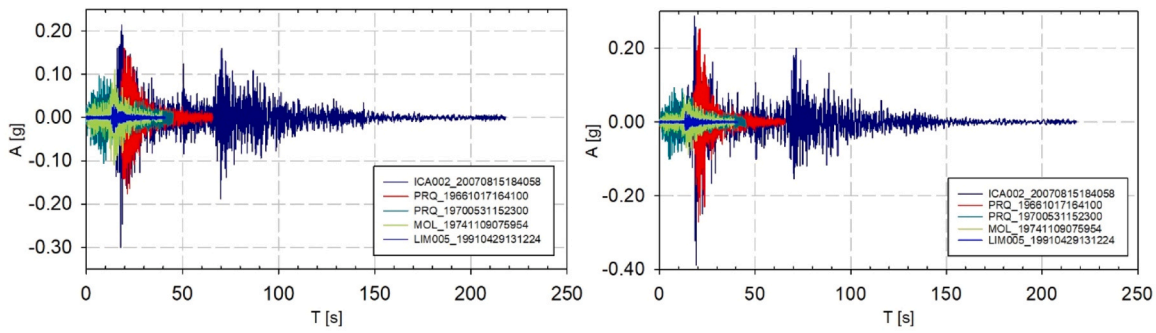
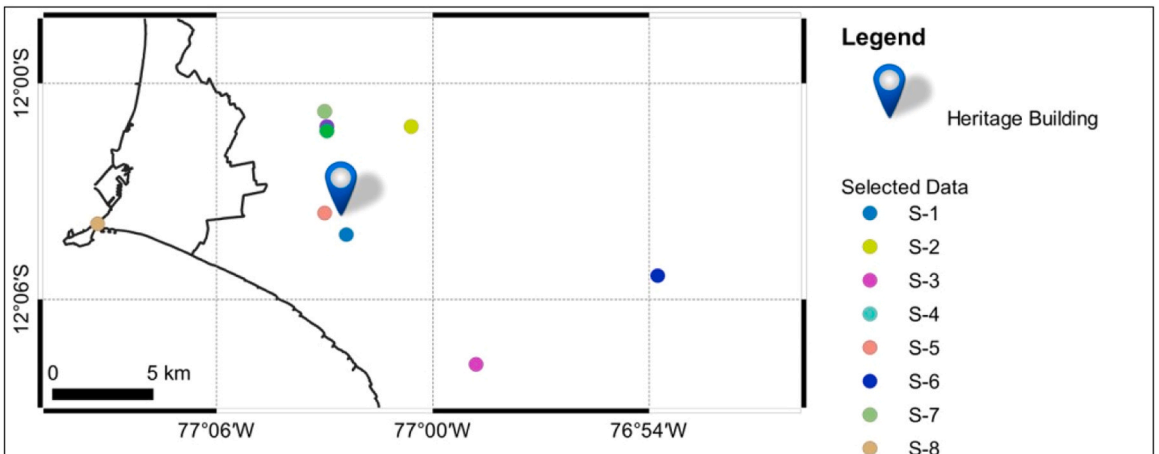


Fig. 8. Time-history seismic ground motion records selected grouped by direction N-W (right) and E-W (left).



Selected Data Table

ID	Station ID	EE Date	PGA EW (g)	PGA NS (g)	PGA UP (g)	Mw	Dist.* (km)
S-1	PRQ	10/17/1966	0.198	0.273	0.031	8	222.06
S-10	LIM-UNI1	1/30/2012	0.006	0.005	0.004	6.4	274.26
S-2	ZAR	1/5/1974	0.142	0.159	0.079	6.6	89.61
S-3	SCO	10/3/1974	0.196	0.211	0.128	7.7	59.3
S-4	MOL	11/9/1974	0.119	0.095	0.042	7.2	78.24
S-5	LIM005	4/29/1991	0.036	0.005	0.003	6.1	97.86
S-6	LIM004	8/15/2007	0.08	0.07	0.058	8	165.89
S-7	LIM001	8/15/2007	0.075	0.007	0.003	8	165.89
S-8	CAL001	8/15/2007	0.103	0.098	0.032	8	167.38
S-9	LIM002	1/30/2012	0.005	0.005	0.004	6.4	274.26

Table Notes:
EE: Earthquke event
* Distance from EE to Heritage Building

Fig. 9. PGA summary disaggregated by distance to source, magnitude, and event.

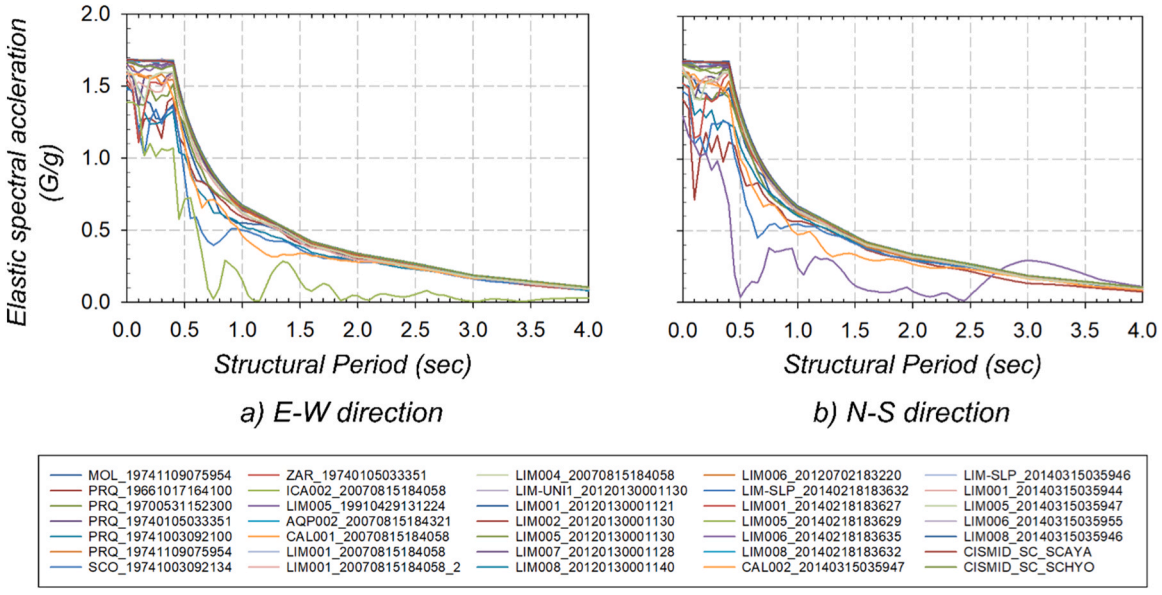


Fig. 10. Elastic response spectrum comparison for selected events.

4.1. Model calibration

A numerical model was set up to capture the proper dynamic response of the heritage building, considered as a mixed structural system with masonry walls and interior reinforced concrete columns as well as reinforced concrete vaults. The calibration process included the development of two different FEM models to idealize the building's response. The first was a multi-degree-of-freedom (MDoF) elastic linear model based on finite elements (i.e., shell and frame elements) implemented in the software SAP2000 [61]. The second was a detailed MDoF non-linear model implemented in the software 3DEC Version 9.0 [49]; for the sake of visualization, Fig. 12 presents screenshots of both. The 3DEC model was constructed using deformable blocks with a finite-deformation FEM formulation. The structure was discretized into heterogeneous zones with distinct material properties and not treated as a homogeneous or rigid discrete system. The "block" element represents solid or volume elements with interactions at joints (i.e., contact zones) that are divided into tetrahedral zones to allow deformability [49]. The

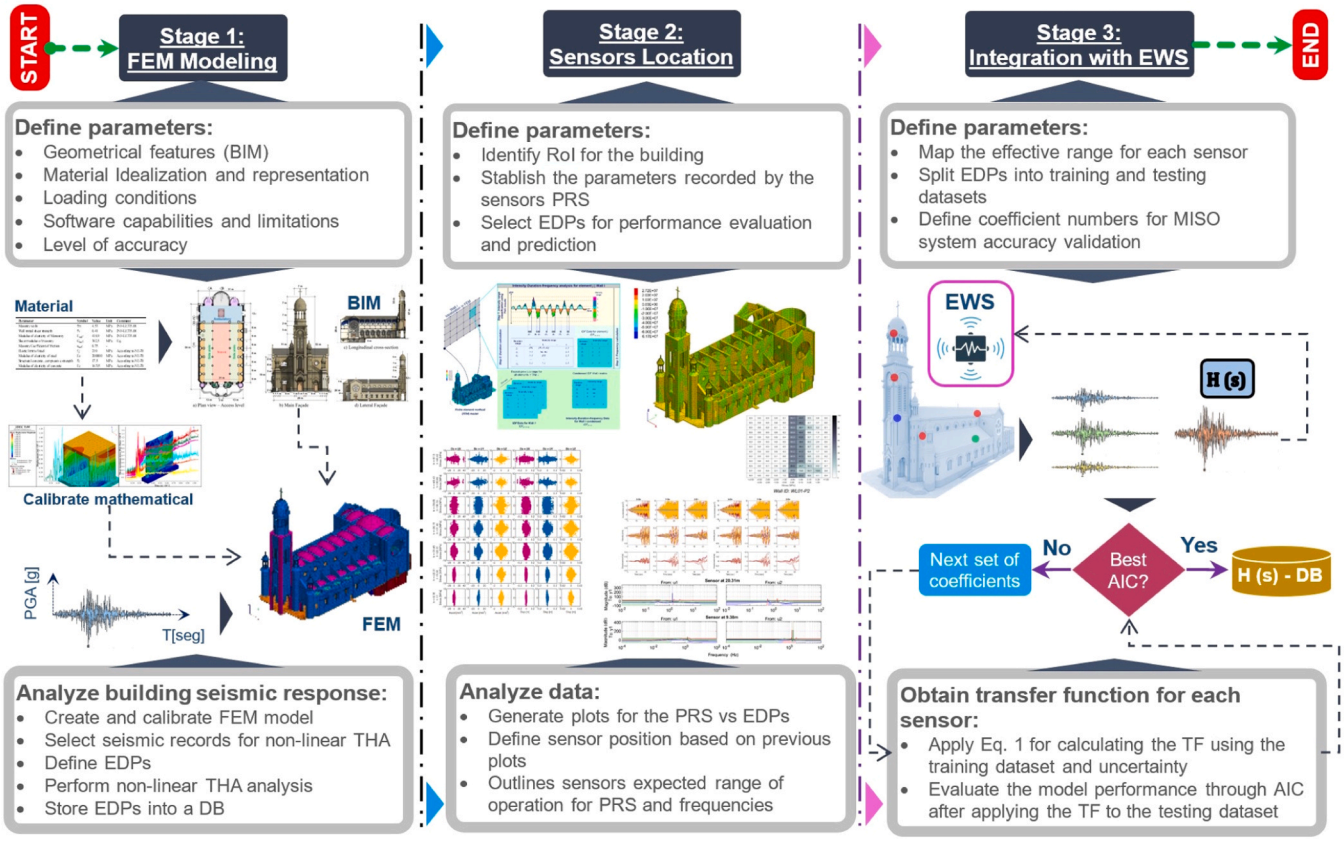


Fig. 11. Methodological approach for defining the inputs of the EWS using the DT model.

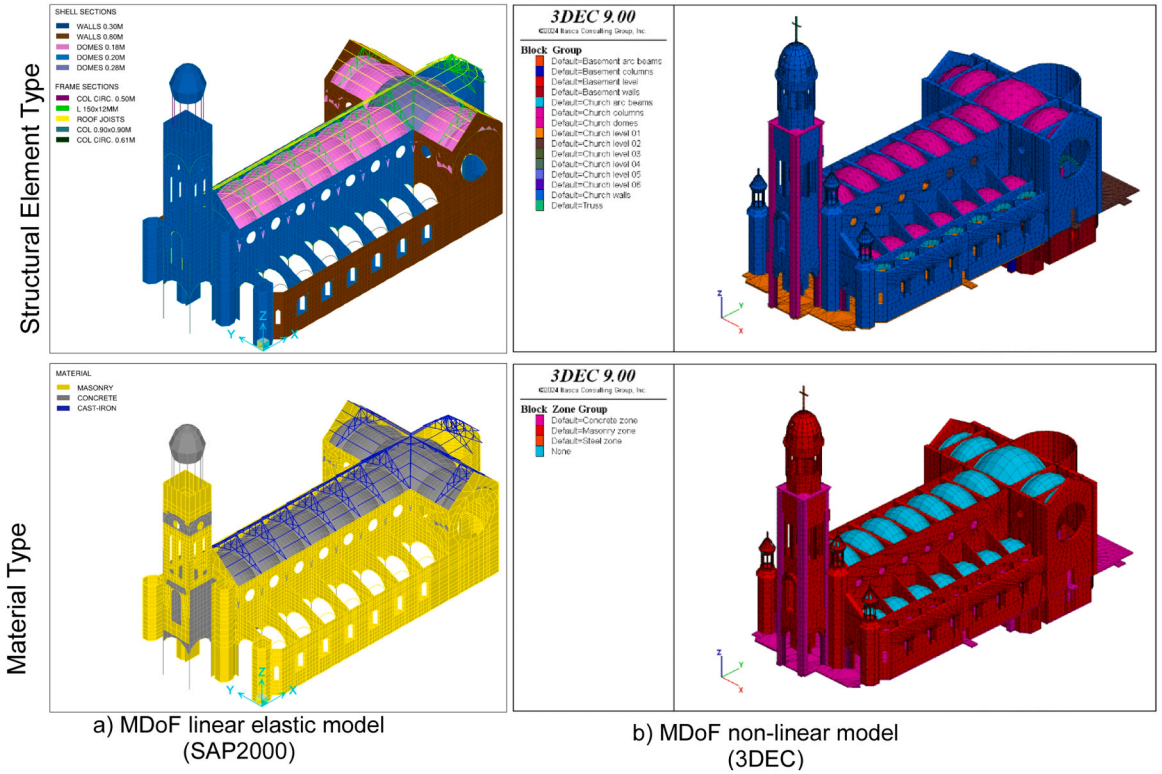


Fig. 12. Axonometric views of the models developed.

interaction between contact zones and joints was set using the Mohr-Coulomb model. As a final remark, the linear and non-linear elastic behavior of the materials was based on the current state-of-the-art [3,22].

The linear-elastic MDoF model developed in SAP2000 was used as a preliminary benchmark to calibrate global dynamic behaviors such as mass distribution, stiffness, and modal characteristics of the more detailed non-linear FEM model implemented in 3DEC. The calibration process involved matching the fundamental vibration periods and modal shapes obtained from both models, ensuring consistency in mass participation factors and overall structural behavior. Adjustments were made to the material stiffness and boundary conditions in the 3DEC model to align its dynamic response with the results of the SAP2000 model. Although 3DEC is capable of both FEM and DEM analysis, the model used in this study was developed entirely using FEM-based deformable block elements with contact interface behavior, and no distinct element formulation was applied.

In the next phase, the BIM creation required an extra-mile effort before migrating the model into the final software, despite the numerous bridging BIM -FEM plugins available in the market. Some challenges included unifying edges and completing missing boundary conditions between the multiple elements, and then properly setting assumptions, capabilities, desired accuracy levels, and limitations related to the FEM.

4.2. Modal analysis results

The dynamic response of the building was assessed by means of a modal analysis, applied to the MDoF model (i.e., model 1 - linear elastic model). An eigenvalue analysis was conducted for a total of 220 vibration modes, which achieved at least 80 % of mass participation in each main direction (i.e., longitudinal (UX) and transversal (UY) direction. The seismic mass of the structure, as derived from the FEM models, is approximately 21,400 kN, which corresponds to a total structural mass of about 2180 metric tons. This value was independently verified through a manual estimation based on volumetric measurements extracted from the BIM model and archival material density data for masonry, concrete, and steel elements. The comparison between both methods yielded a discrepancy of less than 5 %, confirming the reliability of the modeled seismic mass. Fig. 13 displays the response for the first six modes of vibration extracted from the SAP2000 linear-elastic model, illustrating both translational and torsional behavior along the main structural axes, while Table 3 contains the mass participation factors and structural periods for the fundamental modes in the three translational and rotational directions. It is important to note that the structural periods differ by less than 5 % between the models. Finally, the fundamental structural periods in the longitudinal (UX) and transverse (UY) directions of the building are 0.39 sec and 0.38 sec, respectively (refer to modes 3 and 4 in Table 3).

4.3. Time history analysis results

The building's dynamic response was evaluated using THA after validating the detailed model (i.e., model 2 – non-linear elastic model), by applying the set of seismic ground motions selected in Section 3.1. The damage to the structural components was evaluated by means of a comparison between their capacity and demands (unreduced and unfactored). In the case of the masonry walls, concrete columns, and cast-iron trusses, the unreduced or nominal capacity was used. Moreover, horizontal and vertical displacement limits were considered as part of the performance-based assessment conducted on the heritage building. Fig. 14 displays the results for the most critical seismic ground motion record, highlighting the axial stresses and shear force demands acting on the masonry walls under dynamic excitation.

The vast volume of data generated by the THA implied a complex and time-consuming processing task. To address this challenge, an innovative approach systematizes the data through intensity-duration-frequency (IDF) plots, organized according to key structural elements, such as wall typologies. The first step consists in defining the structural element under consideration; subsequently, the intensity is identified and classified into a discretized range of relevant parameters for evaluation (i.e., stresses, displacements, velocities, or accelerations); the discretization of these ranges is based on control points according to the specific objectives expected by the authors from the THA. Once this has been defined, the duration is calculated based on the time a given

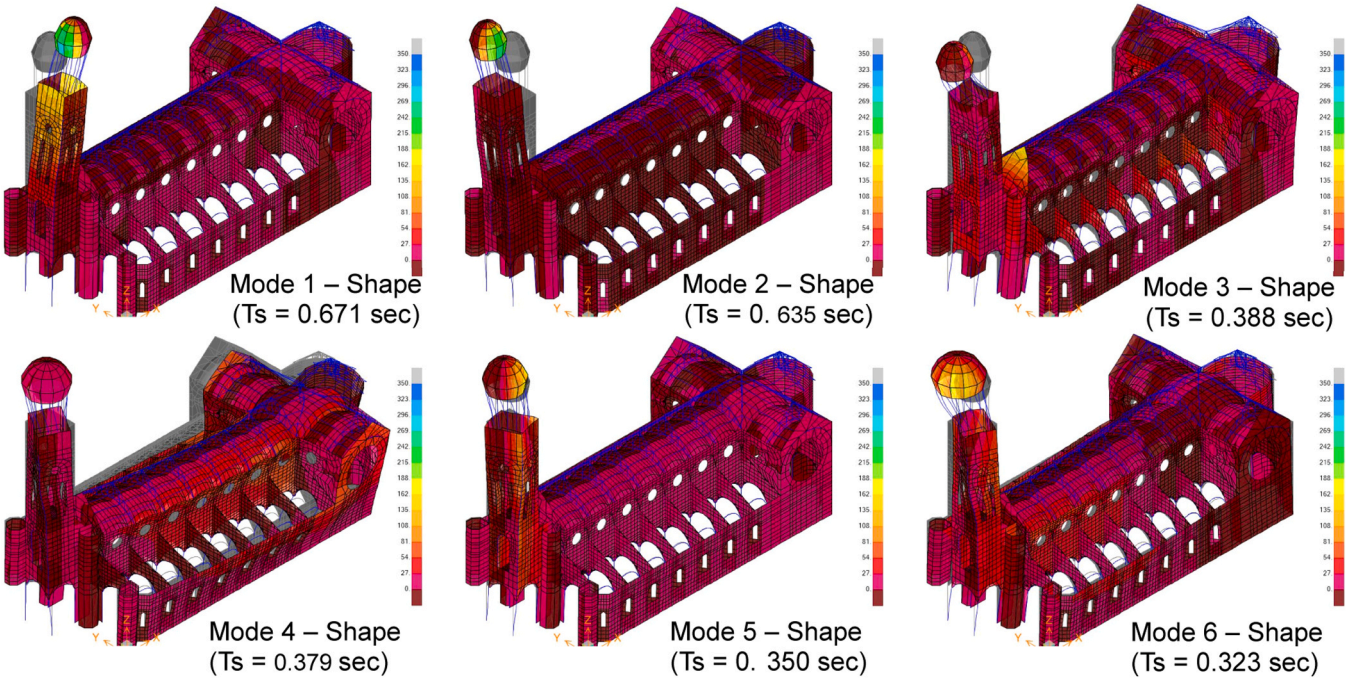
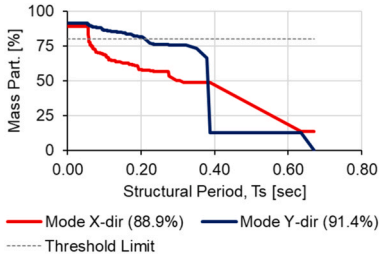


Fig. 13. First six vibration modes from the linear-elastic SAP2000 model. Modes include translational (Modes 1, 2, 4) and torsional or coupled (Modes 3, 5, 6) responses. *Each mode is labeled with its corresponding structural vibration period (in seconds).*

Table 3
Summary of modal response results for the detailed linear elastic model.

Transl. Direction	Mode	Ts sec	Mass Part. %	Rot. Direction	Mode	Ts sec	Mass Part. %
UX	3	0.388	35.4	RX	2	0.635	21.3
	1	0.671	13.4		216	0.074	1.5
	11	0.274	3.2		21	0.208	1.1
UY	4	0.379	53.6	RY	1	0.671	7.8
	2	0.635	12.7		166	0.090	3.7
	5	0.350	6.8		189	0.083	2.9
UZ	189	0.083	4.8	RZ	6	0.323	22.1
	186	0.084	3.5		2	0.635	18.8
	192	0.081	2.6		4	0.379	9.8

Accumulated modal mass participation



intensity is sustained within the previously defined range. Finally, the frequency refers to the number of sub-elements within the primary element that exhibit behaviors within a specific range of intensity and duration. This approach provides a clear and comprehensible representation of complex THA results, easing the analysis using IDF plots. Fig. 15 illustrates the general framework behind the tool.

Following on, Fig. 16 summarizes in-plane and out-of-plane drifts, including the vertical displacement and acceleration for the 3 translational *DoF*; likewise, Fig. 17 displays the stresses in each direction, while Appendix B contains the complete set of results for the walls of the *IDF* plots.

Results confirmed that the maximum vibration-period for the wall under analysis reached 1.3 sec for any of the considered parameters when focusing on "low range intensities", and dropped to almost 0.50 sec for higher intensity ranges; for extreme conditions overpassing the threshold, the duration descended to less than 0.20 sec.

4.4. Comparison between the numerical model and the historical damage reported

The FEM model performance, subjected to a subset of seismic ground motion records (related to the ICA Earthquake event – 08.15.2007), was compared against the aforementioned damage report. In general, it accurately confirmed the crack's distribution over the barred vault reinforced concrete elements (refer to Fig. 18) and corroborated the most vulnerable areas for the real structure. The cracks reported in the numerical model corresponded to the regions of the reinforced concrete vaults that surpassed the concrete tensile stress threshold (i.e., modulus of rupture for concrete with a value of 2.84 MPa). Some isolated zones also exceeded this limit in the FEM, even though the damage report *DB* did not report the same.

4.5. Recommended control points for implementing the digital-twin model

This stage aims to guarantee real-time data acquisition to link the physical and virtual worlds by addressing the optimal sensor location. The distribution of the control points where the in-field sensors were meant to be installed was based on the identification of the highly sensitive regions prone to damage by seismic earthquakes, as well as those of primary interest that will trigger the EWS. Their final location accounts for the seismic structural response acknowledged by the numerical models, and the building's performance observed from past earthquake events.

During this process, a primer analysis for the masonry walls consisted of tracking the normal to bed joints axial stress as an EDP. In this case, only the main structural walls were considered, as highlighted in Fig. 19. Fig. 20 illustrates the EDP intensity-frequency results from the THA for the main wall in the longitudinal direction (i.e., Wall IDs WL01-P1 and WL01-P2), discretized by height ranges varying from 0 to 24 m, with a 2-meter delta. Appendix C contains the complete set of figures for the other walls. Finally, Fig. 21 presents the summary of the vertical axial stress intensity-frequency results for all the piers and the whole set of seismic ground motions.

From Fig. 20 and Fig. 21, it is observed that around 50 % of the walls sustained a vertical axial stress between −0.50 and 0.26 MPa, confirming they were subjected to considerable compression and tensile stress. In addition, a small percentage of the longitudinal piers reached values greater than 1.00 MPa (i.e., around 0.4 %), while for the transverse ones, this value increased up to 2.6 % of the area exceeding this threshold. Conversely, compression stresses greater than 0.50 MPa occurred for a maximum of 2 % and 6 % of the wall area for longitudinal and transverse walls, respectively.

Then, four regions of interest (RoI) were established based on previous results for mapping the masonry wall performance at different range height levels (see Fig. 22). It includes the main tower, two portions of the masonry walls in the central nave extending one-half of the distance between columns in the longitudinal direction, and the outer lateral walls in the ambulatory region. EDPs plots, including displacement and acceleration vs axial stresses (i.e., normal to bed joints direction), were analyzed to characterize the wall's performance using the results obtained from the THA for the non-linear MDoF model (i.e., model 2). Fig. 23 depicts the results of these RoI covering the main tower (i.e., RoI #1) while the central nave and ambulatory region are provided in Appendix D.

From Fig. 23 it is observed that displacement and acceleration show an increase with the height level, where their behavior can be grouped into three levels, i.e., from 7 to 20.31 m, 20.31–39.65 m, and 39.65–51.5 m. Likewise, stresses are higher at lower levels; nonetheless, the last two top levels display an increase in their values. In the case of Level 6, this increase is related to the bell weight, which is supported by it. Similar trends were depicted for the other RoI on the inner and outer walls. As a result, Fig. 24 illustrates the suggested points for positioning the sensors at each RoI identified.

4.6. Recommended transfer function for converting movement time-histories to stresses

The final stage provides an overview of the expected intensity ranges of movement, useful to determine the sensors' requirements and define the transfer functions to forecast the axial stresses on masonry walls. On the one hand, EDPs results from the virtual world (i.e., numerical model 2) for the control points defined in the previous stage (Section 4.5) were used as input for analyzing their response while considering the three components of movement, including acceleration, velocity, and displacement. Fig. 25 shows the results for control points "Group 1a" and "Group 1b" (see Fig. 24), in terms of displacement, velocity, and acceleration EDPs for the first 25 s of the THA related to the ICA earthquake (i.e., ground motion ID ICA002_20070815184058); notice that timeseries are reported for windows of less than 0.01 s, and results are grouped into delta time intervals related to 1.0 s and a fundamental period of 0.39 s (Table 3).

Furthermore, Fig. 26 displays the displacement-THA in terms of an

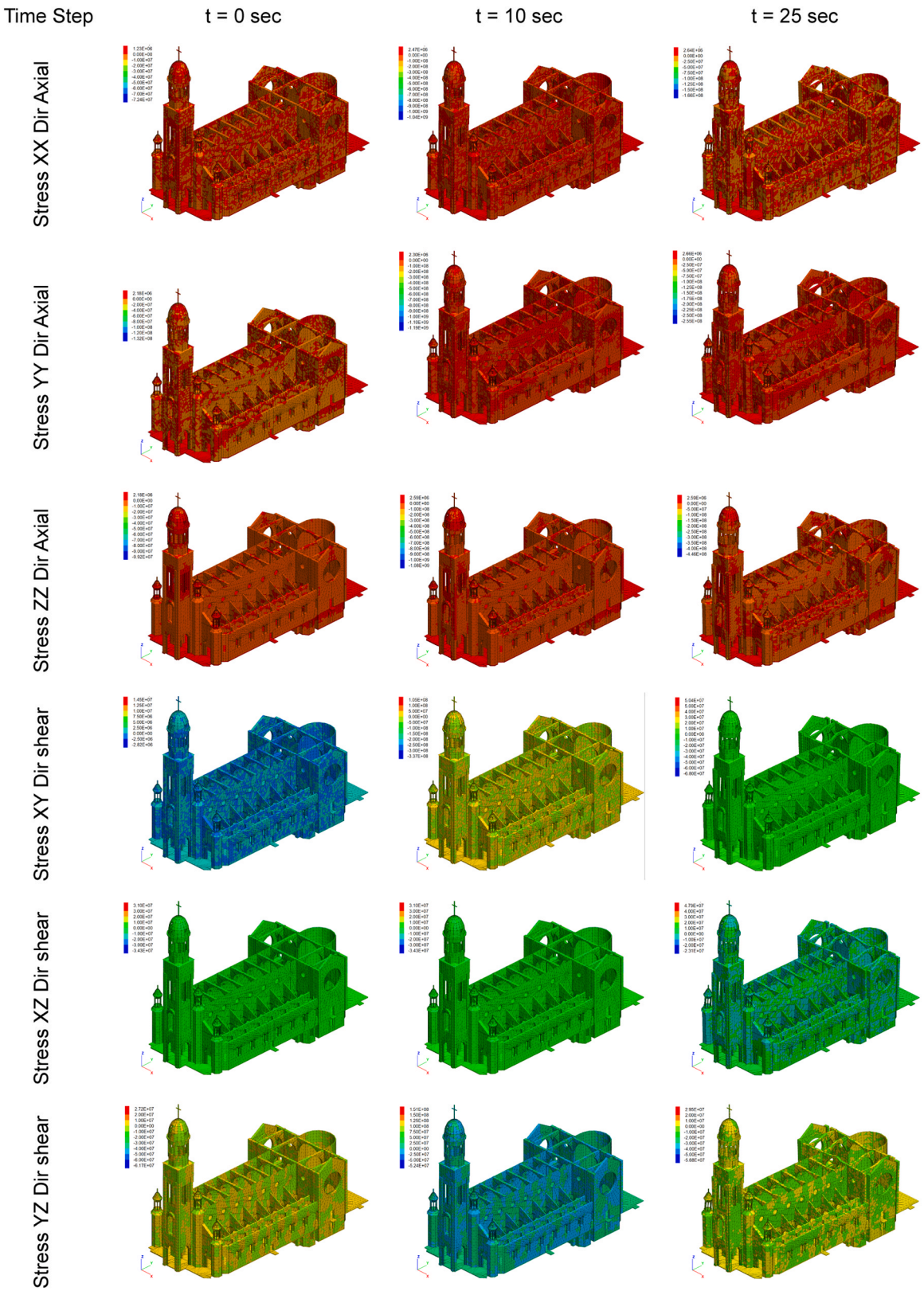


Fig. 14. Time-history response analysis for the most critical ground motion record in masonry walls stresses (units in Pa).

intensity-frequency plot to easily identify the controlling ranges when considering the whole set of assessed ground motions. Other observations indicate displacements reaching extreme values up to 0.20 m and 0.40 m in the x- and y-directions, respectively (see Fig. 26). For the former, 70–80 % of the values fall in that range, while for the latter, 60 % of the outputs range between −3.0 cm and 1.0 cm; Z-direction depicts no clear trend.

Alternatively, the prediction of masonry wall stresses was performed using a nonlinear Multiple-Input-Single-Output (MISO) system based on deep learning (DL) techniques. Specifically, a Long Short-Term Memory

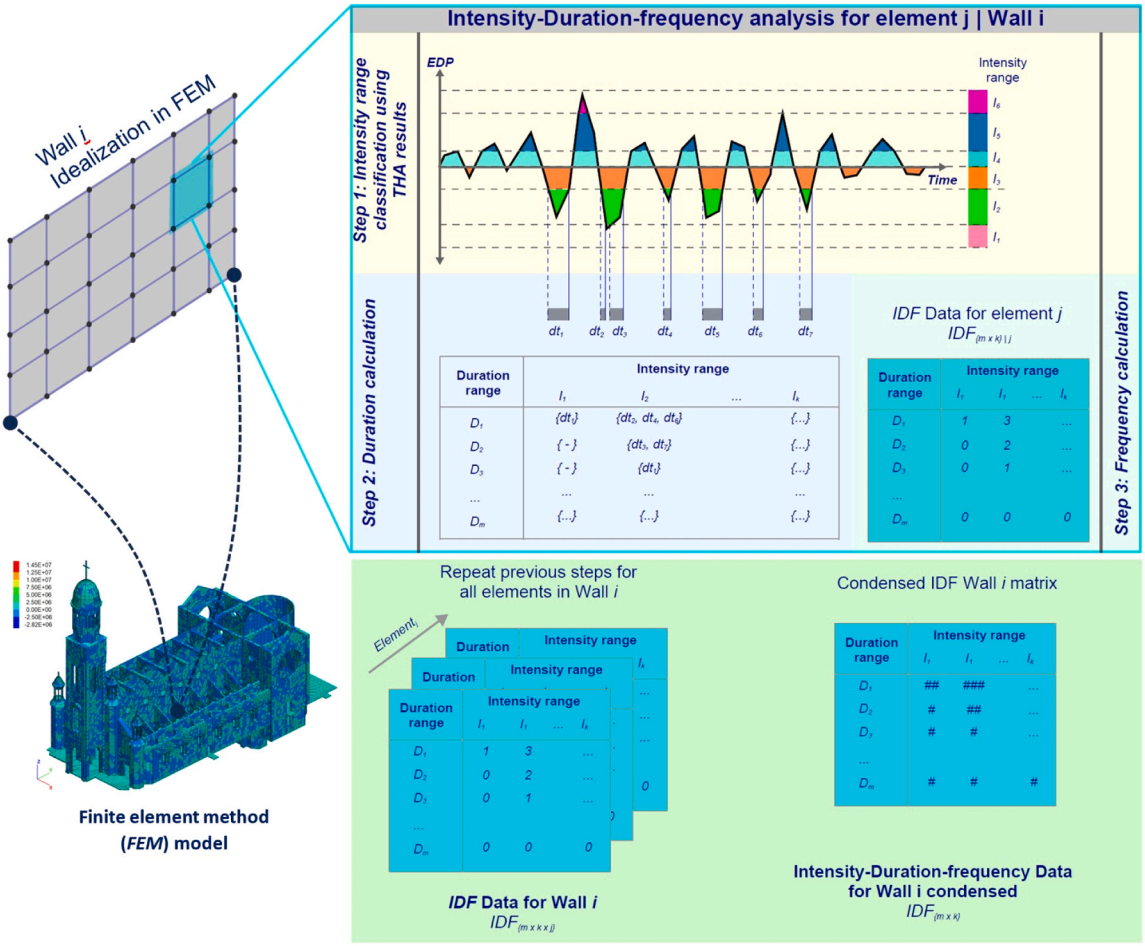


Fig. 15. Schematic flow chart for calculation of *IDF* plots.

(LSTM) model, a type of Recurrent Neural Network (RNN), was selected. This model is well-suited for sequence-to-sequence regression tasks and can effectively leverage past time steps [62]. Input signals for acceleration, velocities, and displacement were analyzed, considering two movement direction scenarios: i) horizontal movement (UX and UY) and ii) three translational directions (UX, UY, and UZ). The output signal was defined as the normal axial stress at the bed joint. Each dataset was subsequently divided into training (70 %), validation (15 %), and testing (15 %) sets. Furthermore, each time series was segmented using a time window of 4.0 s with an overlap of 0.5 s.

The entire process was implemented and solved using a MATLAB-based workflow [63] (as detailed in Appendix E). This methodology was applied to the main tower at control points "Group 1a" and "Group 1b" (see Fig. 24), excluding the top level (51.42 m height). Fig. 27 illustrates the Root Mean Squared Error (RMSE) performance evolution for the training and validation sets when the LSTM model was trained using acceleration input signals. Finally, Fig. 28 presents the overall performance, in terms of RMSE, for the 24 LSTM models evaluated on the validation and testing sets (resulting from 4 control points x 2 direction scenarios x 3 movement components).

The results illustrated in Fig. 27 show the evolution of RMSE over training iterations for two input signal scenarios. Four sensor heights were analyzed: 9.38 m, 20.31 m, 36.15 m, and 39.65 m. The continuous lines represent training RMSE, while dashed lines indicate validation RMSE. A general trend observed across both cases is the decreasing RMSE over iterations, confirming successful convergence of the LSTM model. Moreover, the model performs significantly better when the sensor is placed at lower heights (i.e., 9.38 m and 20.31 m), where both the training and validation RMSE remain below 0.1. This suggests that signal quality and information content are richer at these locations, likely due to higher energy response concentrations and better observability of local deformation patterns, which align with the nonlinear FEM-predicted damage zones.

Furthermore, Case 2 consistently outperforms Case 1, with reduced RMSE levels across all control points. This confirms the added predictive value of the vertical motion component (UZ) in estimating axial stress—an important insight, as vertical accelerations are typically underestimated in masonry monitoring due to assumptions of high axial stiffness. The results here emphasize that vertical ground motion, while small in magnitude, carries critical information regarding through-thickness cracking and stress redistribution, especially in complex geometries such as towers.

Fig. 28 presents a summarized comparison of RMSE values obtained from the 24 LSTM models applied to validation and testing datasets. The lowest RMSEs are again concentrated in sensors located at 9.38 m and 20.31 m, supporting the findings from Fig. 27. Notably, the RMSE values remain under 0.1 in most test cases, which reflects a high level of generalization and robustness of the LSTM models. However, a slight increase in error variability is seen at the higher elevations (36.15 m and 39.65 m), particularly under Case 1 scenarios. This suggests that elevated regions of the tower present higher complexity in dynamic behavior due to boundary condition interactions, modal coupling, and potentially unmodeled phenomena like torsion or out-of-plane flexure.

In summary, the figures provide empirical support for the effectiveness of LSTM-based models in predicting nonlinear axial stress responses in heritage masonry walls using real-time acceleration, velocity, and displacement inputs. The results emphasize the importance of optimal sensor placement (lower elevations yield better signal fidelity),

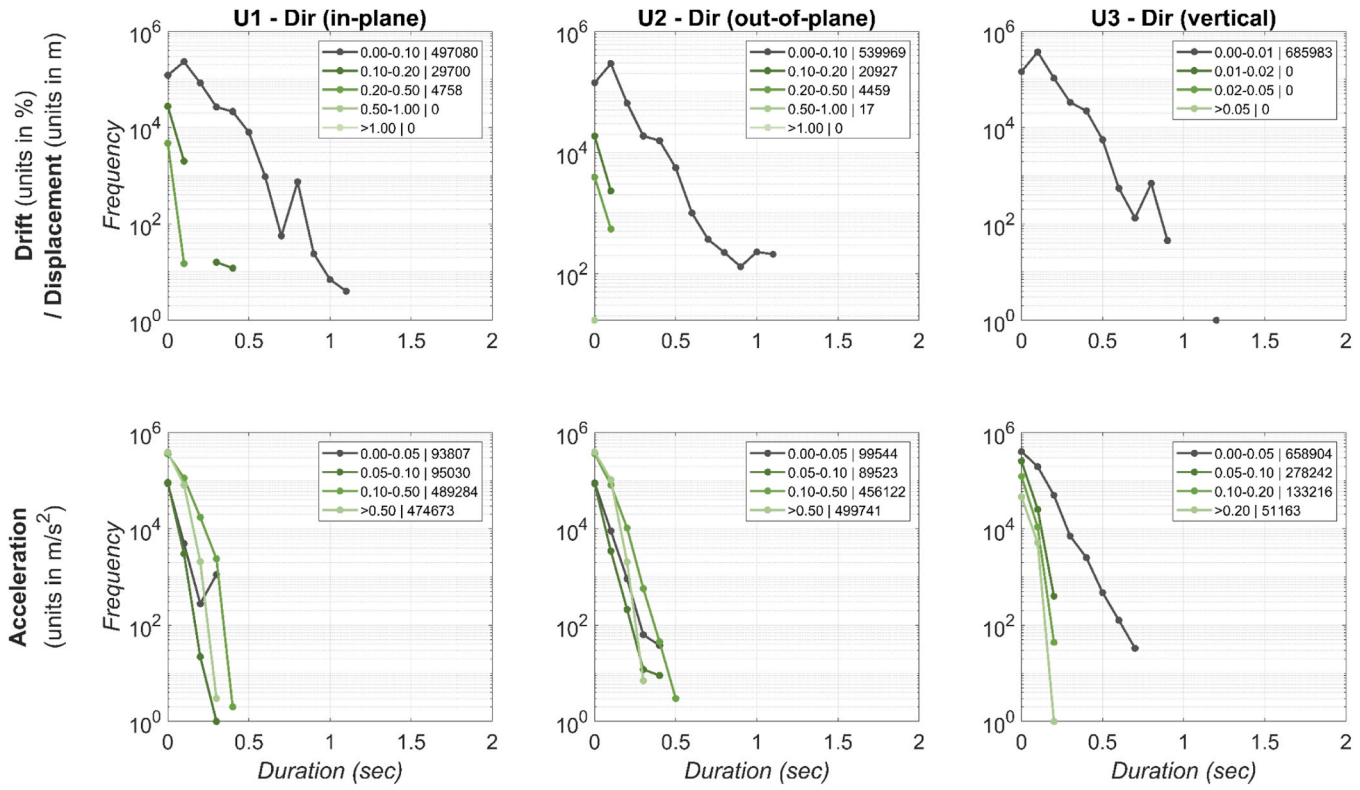


Fig. 16. IDF plots for drift (U1- and U2-Dir), displacement (U3-Dir), and acceleration of a typical inner Longitudinal Wall.

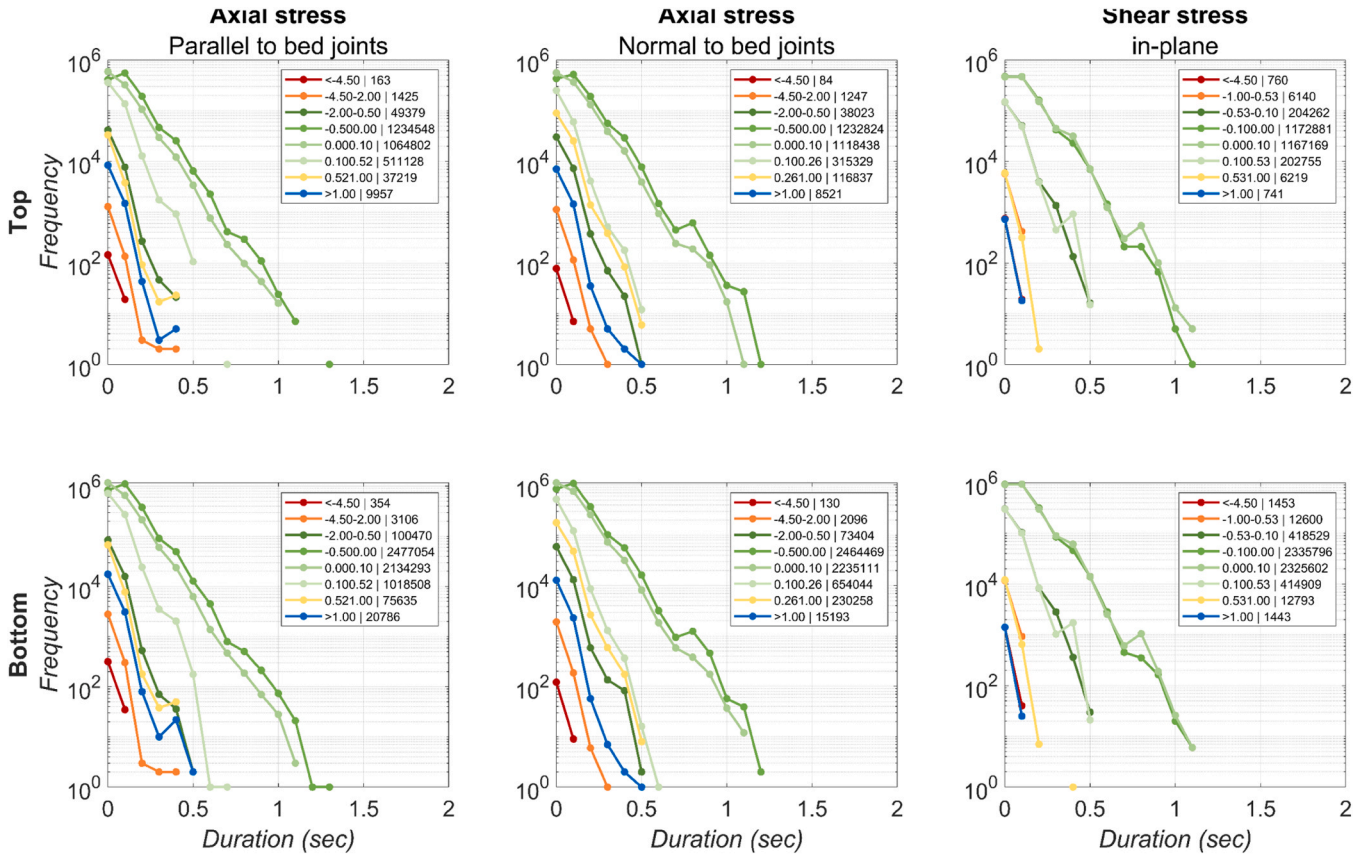


Fig. 17. IDF plots for stresses of a typical inner Longitudinal Wall.

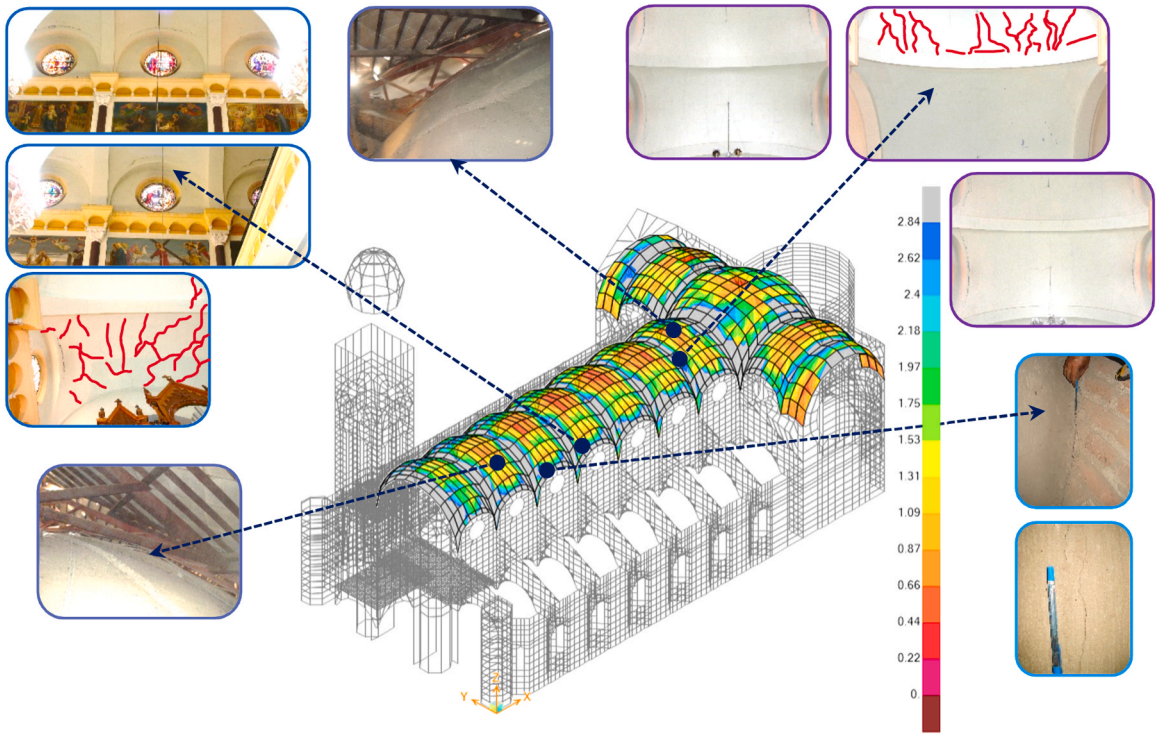
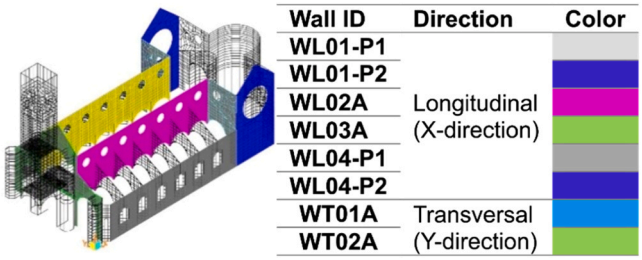


Fig. 18. Comparison between *MDoF THA* results and record of damage reported after the ICA Earthquake event (08–15–2007).



Wall ID	Direction	Color
WL01-P1		
WL01-P2		
WL02A	Longitudinal	
WL03A	(X-direction)	
WL04-P1		
WL04-P2		
WT01A	Transversal	
WT02A	(Y-direction)	

Fig. 19. Masonry wall identification.

the utility of including vertical motion components in SHM-DT integration, and the capability of deep learning to capture complex stress patterns derived from dynamic response histories. These insights reinforce the methodological framework proposed for Intelligent Circular Resilience and the future integration of AI-enhanced digital twins in heritage seismic risk management. Then, tri-axial sensors are meant to be installed on-site in future research to guarantee real-time data acquisition to link the physical and virtual worlds by means of a DT model.

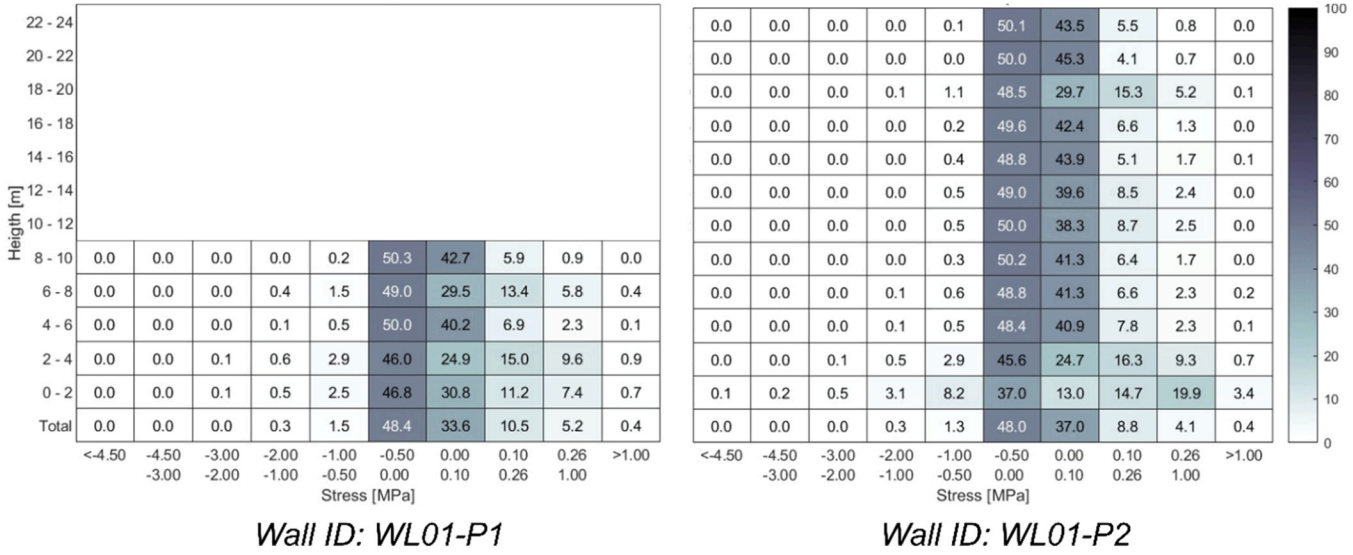


Fig. 20. Vertical axial stress intensity-frequency summary results from *THA* for one of the longitudinal masonry walls.

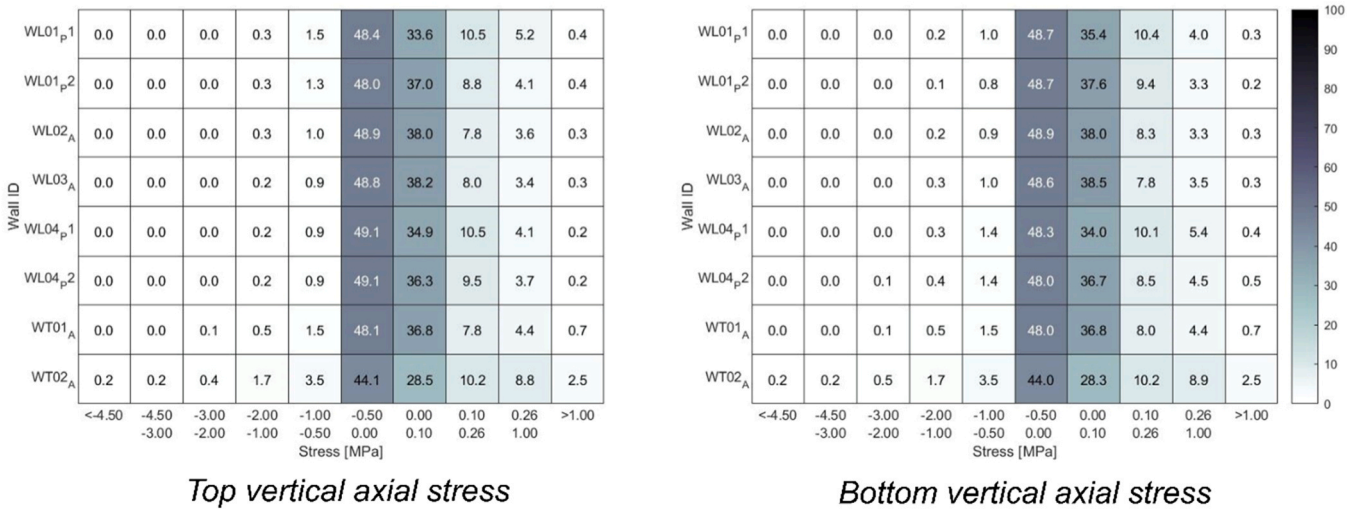


Fig. 21. Vertical axial stress intensity-frequency summary results from *THA* for all masonry walls.

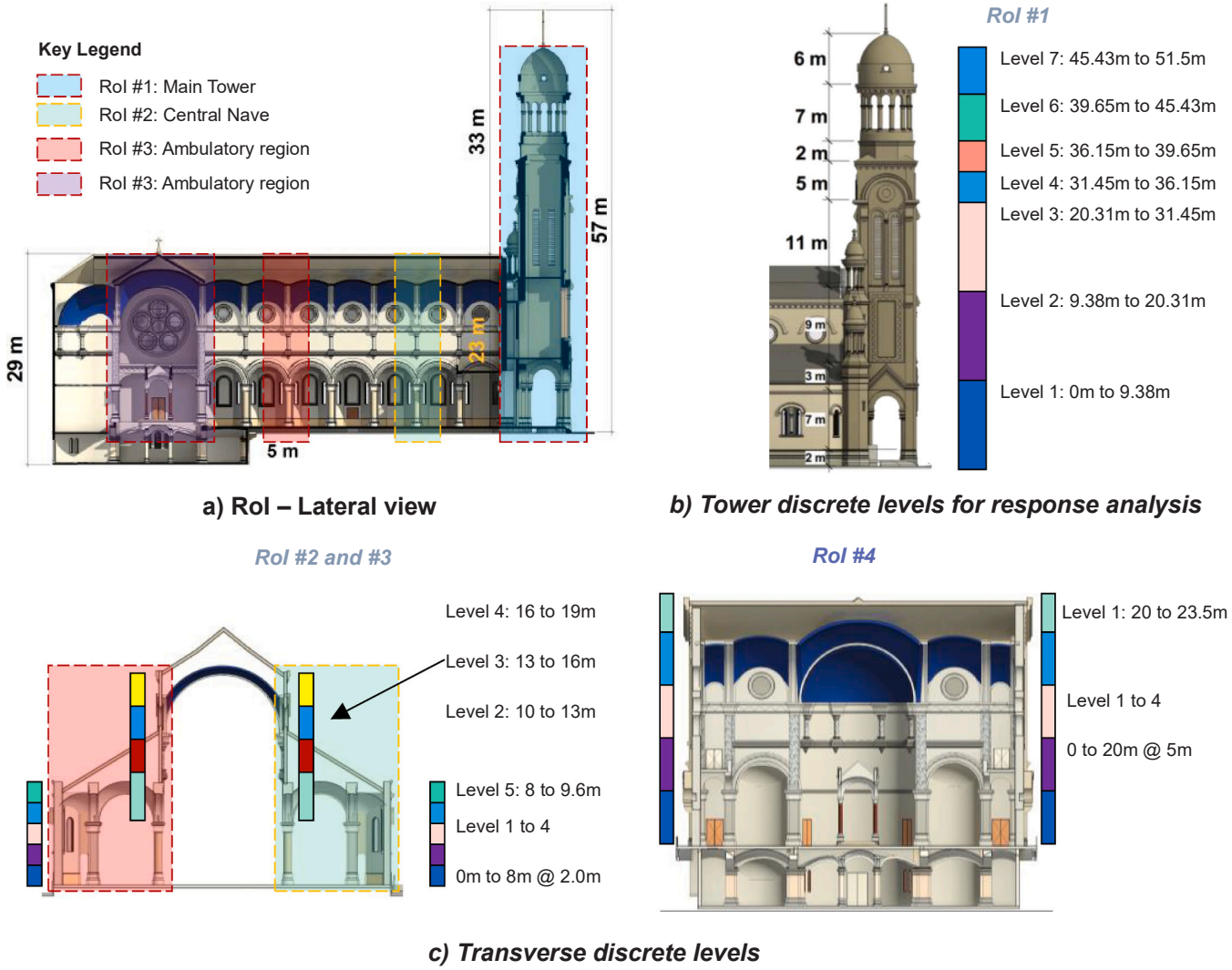


Fig. 22. Regions of Interest for mapping masonry wall performance.

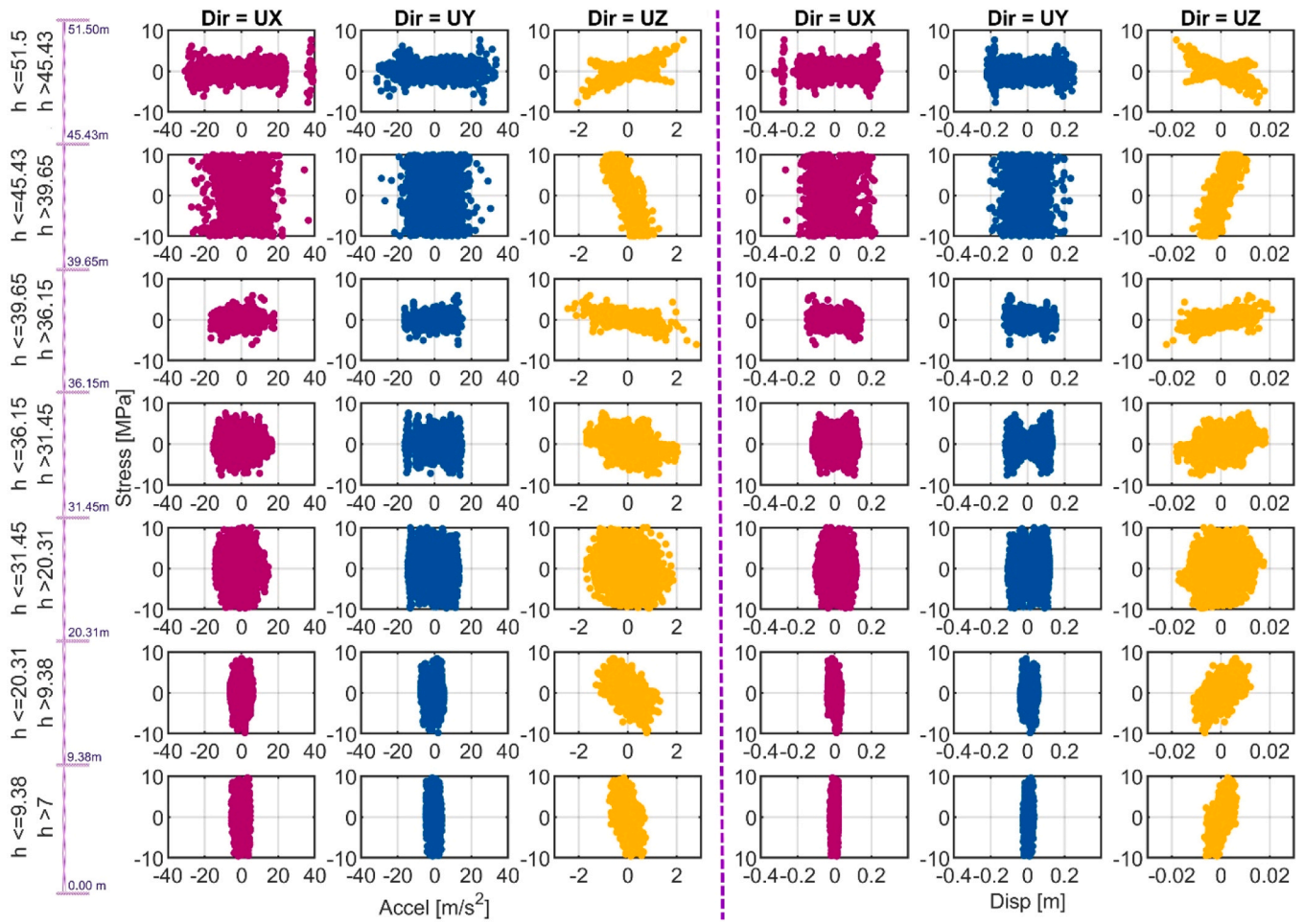


Fig. 23. Masonry wall performance plots for the RoI #1 - main tower. To improve visualization, the performance plots for RoI #1 (main tower) have been separated into individual sub-figures, with enlarged fonts and clearer markers to facilitate comparison across height levels.

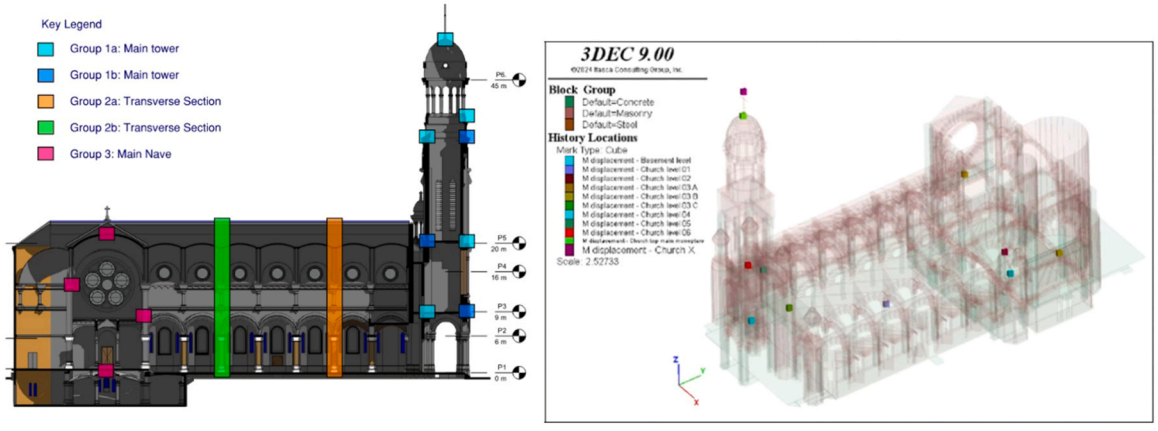


Fig. 24. Recommended points for the installation of sensors: longitudinal transverse view (left), and 3DEC *FEM* model (right).

5. Conclusions

Based on the structural seismic response assessment on an existing heritage building, a methodological approach was proposed by the authors to standardize the ideal distribution of sensors devoted to real-time data feeding of digital twins. This methodology involved the identification of its main characteristics, including materials and geometry, followed by the idealization of the structural response through the development of numerical models. Then, the calibration and validation required a performance-based assessment of the monument, by means of linear and non-linear multi-degree-of-freedom models and a modal-response analysis to identify its dynamic response and behavior under static and dynamic loading conditions. The output allowed the researchers to tackle some of the main challenges faced when dealing with Building Virtualization for the integration of the numerical model into the Digital Twin model.

Subsequently, a time-history analysis was performed for a set of seismic ground motions retrieved from real events recorded at different

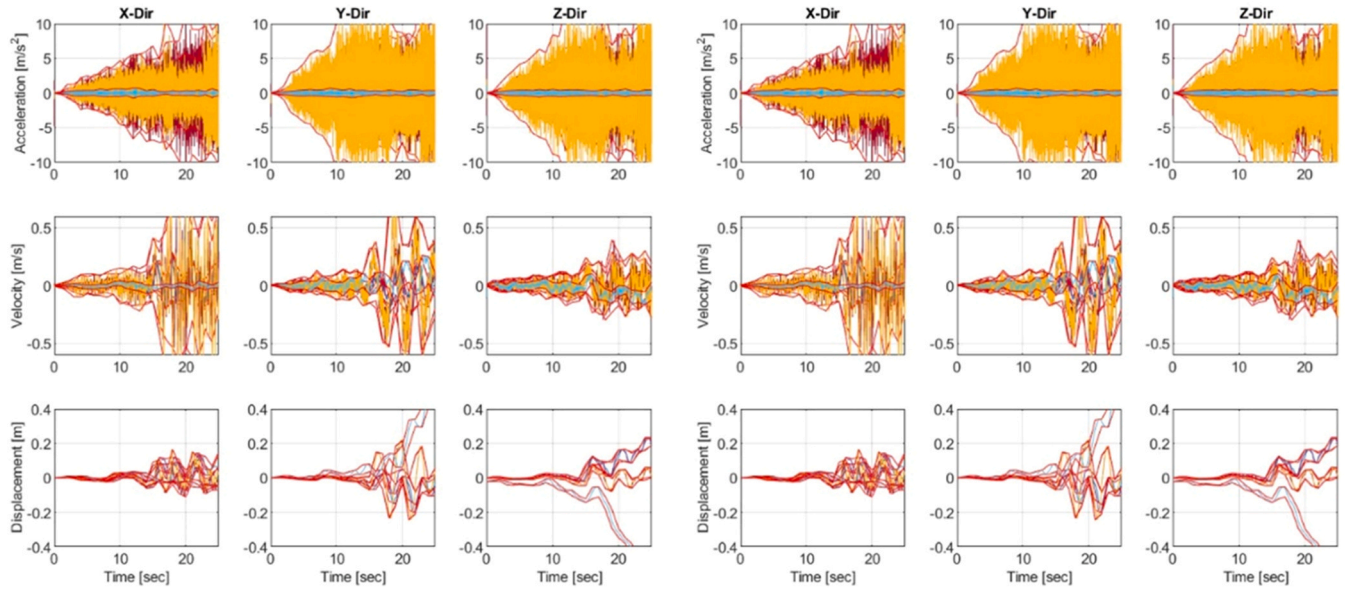


Fig. 25. Summary of EDPs form THA for ICA earthquake event (i.e., records ID ICA002_20070815184058).

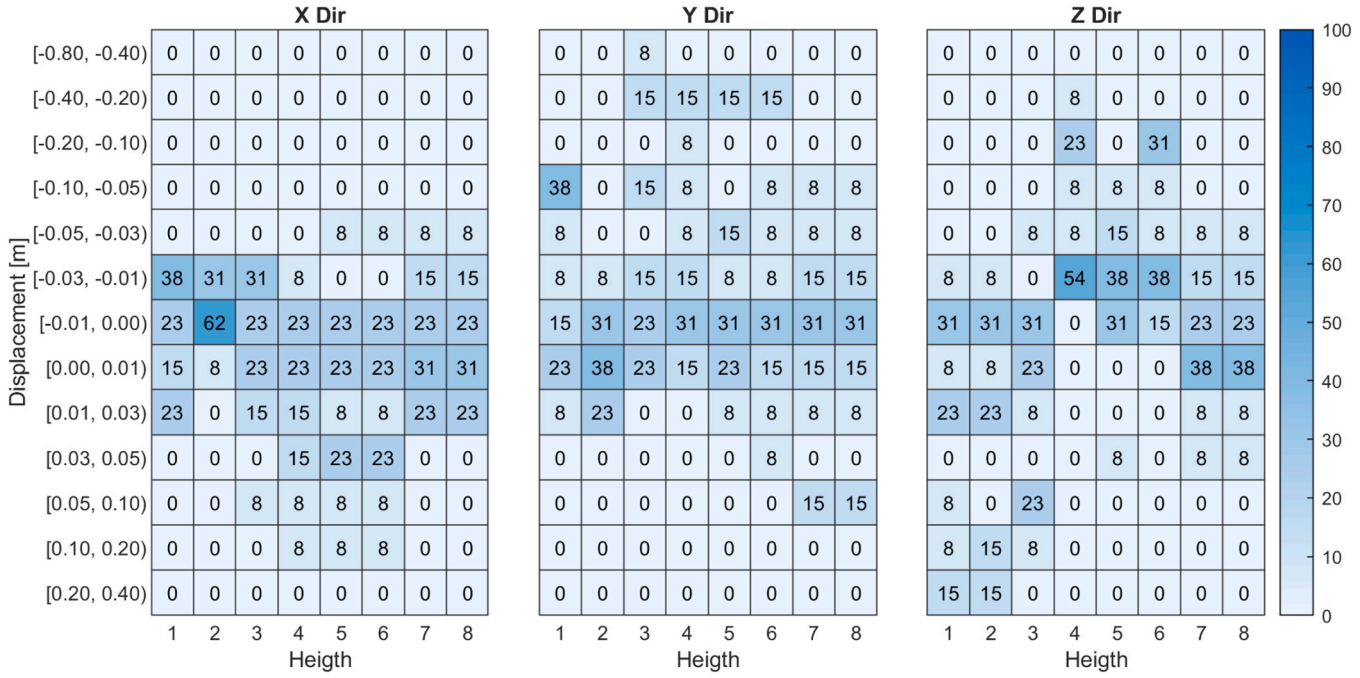


Fig. 26. Summary of displacement intensity-frequency.

seismic stations within the vicinity of the zone of interest. The results showed a trend where most of the records did not induce significant damage to the structural components of the building and confirmed its healthy condition (which is aligned with the historical damage reports). In the case of the few records that triggered damage, the cracking patterns also matched with the outputs from the numerical models and documentation (highlighting the most vulnerable areas of the building). For these specific cases, special attention needs to be given, and a deeper sensitivity analysis should be conducted to enhance the updating of the numerical model due to uncertainties associated with the modeling process related to mechanical material properties.

Furthermore, the authors proposed a novel way for visualizing the vast amount of data generated by the time-history analysis through an intensity-duration-frequency (IDF) plot. This novel proposal was also used to identify the intensity ranges of the engineering demand parameters that later would be correlated with the selection of sensors placed in the field, and the expected range of operation. In this study, the maximum sustained duration for any of the analyzed EDPs reached values up to 1.3 sec, 0.5 sec, and 0.2 sec at low, high, and extreme intensity ranges, respectively. Similarly, a set of 25 control points was defined and validated based on the findings from the numerical model, guiding the installation of the in-field sensors that will interconnect the physical and virtual worlds. The validation process depicted critical displacement values up to 0.40 m in the horizontal direction, which were, on average, way lower (i.e., 3.0 cm for 60 % of the entire duration of the seismic records).

The numerical analysis also provided valuable key points regarding the sensor's characteristics for their future selection. On one hand, the

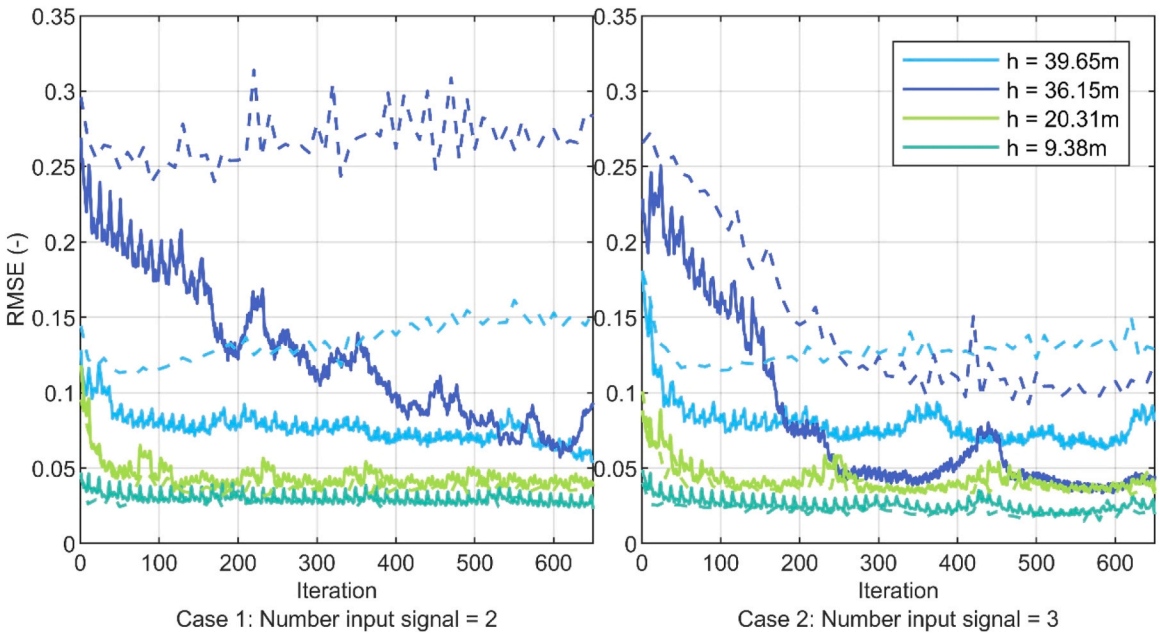
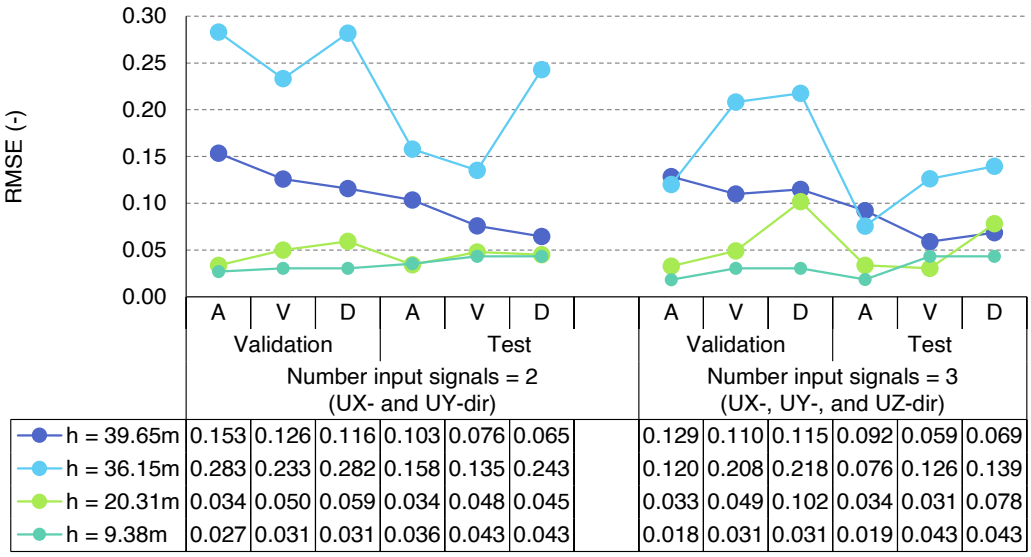


Fig. 27. RMSE performance for acceleration sets.



	Number input signals = 2 (UX- and UY-dir)						Number input signals = 3 (UX-, UY-, and UZ-dir)					
	Validation			Test			Validation			Test		
	A	V	D	A	V	D	A	V	D	A	V	D
h = 39.65m	0.153	0.126	0.116	0.103	0.076	0.065	0.129	0.110	0.115	0.092	0.059	0.069
h = 36.15m	0.283	0.233	0.282	0.158	0.135	0.243	0.120	0.208	0.218	0.076	0.126	0.139
h = 20.31m	0.034	0.050	0.059	0.034	0.048	0.045	0.033	0.049	0.102	0.034	0.031	0.078
h = 9.38m	0.027	0.031	0.031	0.036	0.043	0.043	0.018	0.031	0.031	0.019	0.043	0.043

Fig. 28. Overall RMSE LSTM performance for all sets analyzed.

frequency range for sensors varies from 1.5 to 25 Hz, where the upper limit represents the structural period at which 80 % of mass participation is achieved, while the lower bound corresponds to the longest structural period; all these values are retrieved from the modal analysis. On the other hand, the maximum expected acceleration is around 1.40 g/G, and velocities of up to 0.80 m/sec. Likewise, tri-axial sensors are recommended based on the performance depicted by the MISO model implemented through the DL approach (i.e., LSTM and RNN techniques).

Once the bridge between reality and virtuality is properly working, the numerical model will be integrated into the digital twin to enhance the understanding of seismic behavior, structural response, and the precise identification of optimal points for sensor placement in the field. Further research based on experimental models is still pending to validate the recommended control points and ensure the proper installation of the sensors. While the current validation relies on historical performance and analytical consistency, future research will include sensor-based dynamic monitoring to directly calibrate the numerical model through real-time experimental data, thus closing the feedback loop within the digital twin framework.

It is worth noting that this article solely addresses two phases of the methodology presented in Fig. 1, Building Virtualization, and partially on Data Gathering and Analysis from Sensors. The proposed methodology (Fig. 2 and Fig. 3) can be extended to any civil engineering structure with a proper adjustment, focusing on their virtual representation for further integration with the DT model and sensor devices for SHM.

The complete development of the methodology will include experimental models and sensor implementation, covering phases 3, 4, 5, and 6. The ultimate goal is to create an integrated tool that enhances the understanding of smart cities, utilizing SHM and digital twins as key components of an EWS, within the framework of Intelligent Circular Resilience (ICR) focused on disaster risk management.

CRediT authorship contribution statement

Christiam Angel: Writing – review & editing. **Alvaro Hurtado:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aroquipa Velasquez Hector:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alvaro Hurtado:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hector Aroquipa:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by Socioeconomic Development Camisea Funds (FOCAM) of the National Autonomous University of Tayacaja Daniel Hernández Morillo, Pampas-Tayacaja, Huancavelica, Peru. Also, the authors acknowledge Juan Hurtado, who was in charge of the development of the 3DEC model.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.istruc.2025.110115.

Data availability

Data will be made available on request.

References

[1] 2023. Seismic Evaluation and Retrofit of Existing Buildings. Reston, VA: American Society of Civil Engineers, 2023. doi: 10.1061/9780784416112.
[2] Pitilakis K, Argyroudis S. Systemic seismic vulnerability and risk analysis of urban systems, lifelines and infrastructures. in Seismic Design of Industrial Facilities, 12. Wiesbaden: Springer Fachmedien Wiesbaden; 2014. p. 209–34. https://doi.org/10.1007/978-3-658-02810-7_18.
[3] Clementi F, Gazzani V, Poiani M, Lenci S. Assessment of seismic behaviour of heritage masonry buildings using numerical modelling. J Build Eng 2016;8:29–47. https://doi.org/10.1016/j.jobe.2016.09.005.
[4] Ferrante A, Clementi F, Milani G. Advanced numerical analyses by the Non-Smooth contact dynamics method of an ancient masonry bell tower. Math Methods Appl Sci Sep. 2020;43(13):7706–25. https://doi.org/10.1002/mma.6113.
[5] Pietro C, Veronica F, Marco D, da Porto F. Effectiveness of seismic mitigation strategies for the Italian masonry building heritage. Procedia Struct Integr 2022;44 (2022):1752–9. https://doi.org/10.1016/j.prostr.2023.01.224.
[6] Valente M, Milani G. Damage assessment and partial failure mechanisms activation of historical masonry churches under seismic actions: three case studies in mantua. Eng Fail Anal 2018;92(June):495–519. https://doi.org/10.1016/j.engfailanal.2018.06.017.
[7] Valente M, Milani G. Seismic assessment of historical masonry towers by means of simplified approaches and standard FEM. Constr Build Mater 2016;108:74–104. https://doi.org/10.1016/j.conbuildmat.2016.01.025.
[8] Jiménez-Pacheco J, Quezada R, Calderón-Brito J, Ortega-Guamán E, García H. Characterisation of the built heritage of historic centres oriented to the assessment of its seismic vulnerability: the case of cuenca, Ecuador. Int J Disaster Risk Reduct 2022;71(January). https://doi.org/10.1016/j.ijdrr.2022.102784.
[9] Moropoulou A, Cakmak AS, Lohvyn N. Earthquake resistant construction techniques and materials on byzantine monuments in Kiev. Soil Dyn Earthq Eng Dec. 2000;19(8):603–15. https://doi.org/10.1016/S0267-7261(00)00021-X.
[10] Barbieri G, Biolzi L, Bocciarelli M, Fregonese L, Frigeri A. Assessing the seismic vulnerability of a historical building. Eng Struct 2013;57:523–35. https://doi.org/10.1016/j.engstruct.2013.09.045.
[11] Güllü H, Karabekmez M. Effect of near-fault and far-fault earthquakes on a historical masonry mosque through 3D dynamic soil-structure interaction. Eng Struct Dec. 2017;152:465–92. https://doi.org/10.1016/j.engstruct.2017.09.031.
[12] Batool F, Jabbar Sangi A, Saad Khan M, Islam K. Material characterization of a historical wall by using destructive and non-destructive techniques. Structures 2022;46(ember):1669–78. https://doi.org/10.1016/j.istruc.2022.11.006.
[13] Nežerka V, Slížková Z, Tesárek P, Plachý T, Frankeová D, Petráňová V. Comprehensive study on mechanical properties of lime-based pastes with additions of metakaolin and brick dust. Cem Concr Res 2014;64:17–29. https://doi.org/10.1016/j.cemconres.2014.06.006.
[14] Özmen A, Sayın E. Seismic response of a historical masonry bridge under near and Far-fault ground motions. Period Polytech Civ Eng Apr. 2021. https://doi.org/10.3311/PPci.17832.
[15] Özmen A, Sayın E. Evaluation of material properties of cultural heritage building by destructive and non- destructive testing: malatya Taşhoran church case study. Constr Build Mater 2023;392(March). https://doi.org/10.1016/j.conbuildmat.2023.131693.
[16] Pirchio D, et al. An aggregated non-destructive testing (NDT) framework for the assessment of mechanical properties of unreinforced masonry Italian medieval churches. Constr Build Mater 2022;342(May). https://doi.org/10.1016/j.conbuildmat.2022.128041.
[17] Salazar-Hernández C, Cervantes J, Puy-Alquiza MJ, Miranda R. Conservation of building materials of historic monuments using a hybrid formulation. J Cult Herit 2015;16(2):185–91. https://doi.org/10.1016/j.culher.2014.05.004.
[18] Hernández-Ros O.R.A. 1998. Conservación y restauración del patrimonio arquitectonico. 1998..
[19] Wilkin K. Letter from venice. Hudson Rev 2000;53(1):6–15.
[20] Alarcón M, Soto P, Hernández F, Guindos P. Structural health monitoring of South America's first 6-Story experimental light-frame timber-building by using a low-cost RaspberryShake seismic instrumentation. Eng Struct 2023;275(October 2022). https://doi.org/10.1016/j.engstruct.2022.115278.
[21] Blanco H, Boffill Y, Lombillo I, Villegas L. An integrated structural health monitoring system for determining local/global responses of historic masonry buildings. Struct Control Health Monit 2018;25(8):1–20. https://doi.org/10.1002/stc.2196.
[22] Makoond N, Pelà L, Molins C, Roca P, Alarcón D. Automated data analysis for static structural health monitoring of masonry heritage structures. Struct Control Health Monit 2020;27(10):1–25. https://doi.org/10.1002/stc.2581.
[23] Scuro C, Lamonaca F, Porzio S, Milani G, Olivito RS. Internet of things (IoT) for masonry structural health monitoring (SHM): overview and examples of innovative systems. Constr Build Mater 2021;290:123092. https://doi.org/10.1016/j.conbuildmat.2021.123092.
[24] Soleymani A, Jahangir H, Nehdi ML. Damage detection and monitoring in heritage masonry structures: systematic review. Constr Build Mater 2023;397(June): 132402. https://doi.org/10.1016/j.conbuildmat.2023.132402.
[25] Chaves E, Barontini A, Mendes N, Compán V, Lourenço PB. Methodologies and challenges for optimal sensor placement in historical masonry buildings. Sensors 2023;23(23). https://doi.org/10.3390/s23239304.
[26] Gentile C, Ruccolo A, Canali F. Special issue inspection techniques for damage detection in civil engineering structures Long-term monitoring for the condition-based structural maintenance of the milan cathedral. Constr Build Mater 2019;228: 117101. https://doi.org/10.1016/j.conbuildmat.2019.117101.
[27] Jayawickrema UMN, Herath HMCM, Hettiarachchi NK, Sooriyaarachchi HP, Epaarachchi JA. Fibre-optic sensor and deep learning-based structural health monitoring systems for civil structures: a review. Measurement (Lond) 2022;199 (May):111543. https://doi.org/10.1016/j.measurement.2022.111543.
[28] Baduge SK, et al. Artificial intelligence and smart vision for building and construction 4.0: machine and deep learning methods and applications. Autom Constr 2022;141(June):104440. https://doi.org/10.1016/j.autcon.2022.104440.
[29] Davila Delgado JM, Oyedele L. Digital twins for the built environment: learning from conceptual and process models in manufacturing. Adv Eng Inform 2021;49 (March):101332. https://doi.org/10.1016/j.aei.2021.101332.
[30] Friederich J, Francis DP, Lazarova-Molnar S, Mohamed N. A framework for data-driven digital twins for smart manufacturing. Comput Indus 2022;136:103586. https://doi.org/10.1016/j.compind.2021.103586.
[31] Yoon S. In situ modeling methodologies in building operation: a review. Build Environ 2023;230(December 2022):109982. https://doi.org/10.1016/j.buildenv.2023.109982.
[32] Yoon S. Building digital twinning: data, information, and models. J Build Eng 2023;76(February):107021. https://doi.org/10.1016/j.jobe.2023.107021.
[33] Ding W, Yang H, Yu K, Shu J. Crack detection and quantification for concrete structures using UAV and transformer. Autom Constr 2023;152(April). https://doi.org/10.1016/j.autcon.2023.104929.
[34] Koo J, Yoon S. Simultaneous in-situ calibration for physical and virtual sensors towards digital twin-enabled building operations. Adv Eng Inform 2024;59 (October 2023):102239. https://doi.org/10.1016/j.aei.2023.102239.
[35] Teng S, Chen X, Chen G, Cheng L. Structural damage detection based on transfer learning strategy using digital twins of bridges. Mech Syst Signal Process 2023;191 (February):110160. https://doi.org/10.1016/j.ymssp.2023.110160.
[36] Wang B, Li Z, Xu Z, Sun Z, Tian K. Digital twin modeling for structural strength monitoring via transfer learning-based multi-source data fusion. Mech Syst Signal Process 2023;200(June):110625. https://doi.org/10.1016/j.ymssp.2023.110625.
[37] Dang H, Tatipamula M, Nguyen HX. Cloud-Based digital twinning for structural health monitoring using deep learning. IEEE Trans Ind Inf Jun. 2022;18(6): 3820–30. https://doi.org/10.1109/TII.2021.3115119.
[38] Hunhevicz JJ, Motie M, Hall DM. Digital building twins and blockchain for performance-based (smart) contracts. Autom Constr 2022;133:103981. https://doi.org/10.1016/j.autcon.2021.103981.

[39] Levine NM, Spencer BF. Post-Earthquake building evaluation using UAVs: a BIM-Based digital twin framework. Sensors 2022;22(3). https://doi.org/10.3390/s22030873.
[40] Liu Z, Jiang A, Zhang A, Xing Z, Du X. Intelligent prediction method for operation and maintenance safety of prestressed steel structure based on digital twin technology. Adv Civ Eng 2021;2021. https://doi.org/10.1155/2021/6640198.
[41] Mishra M, Lourenço PB, Ramana GV. Structural health monitoring of civil engineering structures by using the Internet of things: a review. J Build Eng May 2022;48:103954. https://doi.org/10.1016/J.JOBE.2021.103954.
[42] Xu J, Shu X, Qiao P, Li S, Xu J. Developing a digital twin model for monitoring building structural health by combining a building information model and a real-scene 3D model. Measurement 2023;217(May):112955. https://doi.org/10.1016/j.measurement.2023.112955.
[43] Aroquipa H, Hurtado A, Angel C, Aroquipa A, Gamarra A, Almeida Del Savio A. A cost-benefit analysis for the appraisal of social and market prices in the probabilistic seismic risk assessment of building portfolios: a methodology for the evaluation of disaster risk reduction programs. Int J Disaster Risk Reduct 2023;90 (October 2022):103637. https://doi.org/10.1016/j.ijdrr.2023.103637.
[44] Loverdos D, Sarhosis V. Geometrical digital twins of masonry structures for documentation and structural assessment using machine learning. Eng Struct 2023;275(PA):115256. https://doi.org/10.1016/j.engstruct.2022.115256.
[45] Loverdos D, Sarhosis V. Pixel-level block classification and crack detection from 3D reconstruction models of masonry structures using convolutional neural networks. Eng Struct 2024;310(April):118113. https://doi.org/10.1016/j.engstruct.2024.118113.
[46] Loverdos D, Sarhosis V. Image2DEM: a geometrical digital twin generator for the detailed structural analysis of existing masonry infrastructure stock. SoftwareX 2023;22:101323. https://doi.org/10.1016/j.softx.2023.101323.
[47] Loverdos D, Sarhosis V. Automatic image-based brick segmentation and crack detection of masonry walls using machine learning. Autom Constr 2022;140 (March):104389. https://doi.org/10.1016/j.autcon.2022.104389.
[48] Loverdos D, Sarhosis V, Adamopoulos E, Drougkas A. An innovative image processing-based framework for the numerical modelling of cracked masonry structures. Autom Constr 2021;125:103633. https://doi.org/10.1016/j.autcon.2021.103633.
[49] I. Itasca Consulting Group, "3DEC - Three-Dimensional Distinct Element Code," 2024, *Itasca, Minneapolis*: 9.0.
[50] D. Franco Cordova, "Cronologia Basilica de María Auxiliadora," Lima, 2016.
[51] Liu C, Zhang P, Xu X. Literature review of digital twin technologies for civil infrastructure. J Infrastruct Intell Resil Sep. 2023;2(3):100050. https://doi.org/10.1016/j.iintel.2023.100050.
[52] Pantoja-rosero BG, Achanta R, Beyer K. Automation in construction Damage-augmented digital twins towards the automated inspection of buildings. Autom Constr 2023;150(March):104842. https://doi.org/10.1016/j.autcon.2023.104842.
[53] Pantoja-rosero BG, Achanta R, Beyer K. Automation in construction Damage-augmented digital twins towards the automated inspection of buildings. Autom Constr 2023;150(March):104842. https://doi.org/10.1016/j.autcon.2023.104842.
[54] Liu C, Zhang P, Xu X. Literature review of digital twin technologies for civil infrastructure. J Infrastruct Intell Resil Sep. 2023;2(3):100050. https://doi.org/10.1016/j.iintel.2023.100050.
[55] Ponte M, Bento R, Vaz SD. A Multi-Disciplinary approach to the seismic assessment of The National palace of sintra. Int J Archit Herit May 2021;15(5):757–78. https://doi.org/10.1080/15583058.2019.1648587.
[56] Franco Cordova DE. La Basílica de María auxiliadora: Los Salesianos y la arquitectura religiosa en Lima (1900-1963). Lima: Universidad Nacional Mayor de san Marcos; 2016.
[57] Brando G, et al. Structural survey and empirical seismic vulnerability assessment of dwellings in the historical centre of cusco, Peru. Int J Archit Herit Oct. 2021;15 (10):1395–423. https://doi.org/10.1080/15583058.2019.1685022.
[58] Aroquipa H, Hurtado A, Murga C, De La Cruz R, Tarque N. Towards Smart Cities: A Methodological Approach for Implementing Intelligent Circular Resilience in Heritage Buildings through Structural Health Monitoring and Digital-Twins-Part A. Int J Disaster Risk Reduct 2025;128:105749. https://doi.org/10.1016/j.ijdrr.2025.105749.
[59] D.F. Córdova, "Basílica de María Auxiliadora de Lima: Patrimonio cultural de la Nación – Congregación Salesiana del Perú," Salesianos. Accessed: Jul. 11, 2024. [Online]. Available: ⟨https://salesianos.pe/basilica-de-maria-auxiliadora-de-lima-patrimonio-cultural-de-la-nacion/⟩.
[60] Ministerio de Vivienda Construcción y Saneamiento, *Norma Técnica - E.030: Diseño Sismorresistente del Reglamento Nacional de Edificaciones*. 2018.
[61] Computers and Structures Inc., "SAP2000: integrated software for structural analysis and design," 2017, *Berkeley California: CSI*: 20.
[62] "Sequence-to-Sequence Regression Using Deep Learning." Accessed: Oct. 08, 2024. [Online]. Available: ⟨https://mathworks.com/help/deeplearning/ug/sequence-to-sequence-regression-using-deep-learning.html⟩.
[63] Inc. The MathWorks, "MATLAB version R2024b," 2024, *The MathWorks, Inc.* [Online]. Available: ⟨https://www.mathworks.com⟩.